

Funding Instruments and Effort Choices in Higher Education *

Guillem Foucault[†] Juan Pal[‡]

Job Market Paper

This draft: September 26, 2026

Abstract

This paper examines the effects of Tuition-free Higher education (TFHE) policies on student enrollment and academic performance, with a focus on the 2016 Chilean reform that granted tuition-free higher education to students from the lowest five income deciles. Using a difference-in-differences approach, we find that TFHE increased enrollment and persistence in higher education on the eligible but had modest effects on graduation and dropout rates. To disentangle the role of student effort from selection effects, we develop a structural model in which students choose effort levels in response to financial incentives. Our results highlight that TFHE expanded access, in particular for low-achieving students. Despite the removal of academic progress requirements, we found no evidence of weakening performance.

Keywords: Financial aid, Effort choices, Higher Education

JEL Codes: I22, I23, I26

*We acknowledge funding from the French National Research Agency (ANR) under the grant MATCHINEQ - ANR-22-CE26-0005-01 and the European Research Council (ERC) under the grant ED-DYNCHOICE - 101162723.

[†]Toulouse School of Economics. guillem.foucault@tse-fr.eu

[‡]Toulouse School of Economics. juan.pal@tse-fr.eu

1 Introduction

Higher education is a central determinant of long-run earnings, inequality, and intergenerational mobility. Despite decades of expansion in attainment, the returns to a college degree have remained large (Card, 1999; Goldin & Katz, 2008), and access to selective institutions is a powerful lever for upward mobility (Chetty et al., 2020). Yet completion is unequally distributed, and the gap in degree attainment between high and low-income youth has widened in recent decades (Bailey and Dynarski, 2011, Dynarski et al., 2022). In response, governments have invested heavily in policies aimed at lowering financial barriers to college. As of today, more than 40 countries operate some form of tuition-free higher education (TFHE) policy, generally limited to public institutions, and recent debates in the United States over student-debt forgiveness, free community college, and need-based grants reflect a similar emphasis. Public spending on higher education in the OECD has risen accordingly (OECD, 2019), raising the question of how aid should be designed to balance equity, fiscal cost, and educational outcomes.

Higher education policies operate on two margins. On the extensive margin, lowering net tuition draws in marginal students who would otherwise have stayed out. On the intensive margin, the design of aid impacts field choice, persistence and completion among those who do enroll. Traditional instruments combine financial relief with performance requirements to keep incentives aligned. The Pell Grant in the United States is conditioned on satisfactory academic progress, and merit-based scholarships typically require a minimum GPA to renew (Scott-Clayton, 2011, Deming and Dynarski, 2010). Subsidized loans expand access but burden graduates with debt that distorts post-college choices (Rothstein and Rouse, 2011, Sieg and Wang, 2018, Black et al., 2023). TFHE sits at one extreme of this design space, eliminating out-of-pocket tuition for a broad eligible population without imposing progress conditions, and thus removing both a financial barrier and a performance lever simultaneously. The ex-ante effect on educational outcomes is therefore ambiguous. Eliminating tuition draws into the system a less-prepared pool of marginal entrants while, for inframarginal students, it relaxes a binding budget constraint, frees up time previously devoted to part-time work, and reduces dropout after negative income shocks. Removing the performance requirement also impacts selection (drawing in students with a higher cost of effort) while reducing incentives for effort among inframarginals (Scott-Clayton, 2011). Countries with TFHE tend to exhibit lower aggregate graduation rates than comparable countries in which students face out-of-pocket fees, but aggregate measures cannot distinguish compositional changes from behavioral responses during college.

This paper studies the effects of a large-scale TFHE reform on enrollment, persistence, and

academic performance, and develops a framework that disentangles selection from changes in effort. We exploit Chile’s 2016 *Gratuidad* reform, which granted tuition-free higher education to students from the lowest five income deciles in roughly half of universities and vocational centers (including some private institutions), as a natural experiment, and proceed in three steps. First, we implement a difference-in-differences design to estimate effects on enrollment, persistence, dropout, and graduation. Second, we build and estimate a dynamic structural model of enrollment, program choice, and academic effort that endogenizes how students respond to changes in out-of-pocket tuition and to the presence or absence of performance requirements. Third, we use the estimated model to simulate counterfactual designs that re-introduce performance-based conditions on top of the tuition-free benefit, and assess their implications for fiscal cost and educational attainment. Together, the reduced-form and structural exercises decompose aggregate impacts into a composition effect from marginal students induced to enroll and a behavioral effect on the effort choices of infra-marginal students.

The Chilean context offers an ideal setting for this analysis, thanks to both rich administrative data and a well-defined policy intervention. We assemble administrative records that link the universe of national-exam takers to higher education enrollment, scholarship assignment, course-by-course credit completion, dropout, and graduation outcomes. The *Gratuidad* reform is a sharp, large-scale policy change with a clearly defined eligibility rule based on income decile, introduced on top of a well-developed pre-existing aid system that included subsidized loans (the State Guaranteed Loan), need-based scholarships, and a centralized admission platform. Crucially, the pre-existing instruments were all performance-conditional, both at entry (scores at the National Entrance Exam) and during studies (course completion and GPA). *Gratuidad* replaced these conditional schemes with full, unconditional tuition coverage for eligible students. The reform therefore varied both the level of net tuition and the performance conditioning of aid, providing the variation needed to identify selection and effort responses separately.

The difference-in-differences estimates reveal a sizable extensive-margin response. Enrollment among eligible students rose by 3.5 percentage points (p.p.) in 2016 and stabilized at around 10 p.p. from 2017 onward, when the policy was extended to vocational centers. The increase is concentrated among lower-ability students who, prior to the reform, did not qualify for either subsidized loans or merit scholarships. Their enrollment rose by 14 p.p., compared with 5.5 p.p. for students already eligible for the most generous pre-existing aid. We find no evidence of crowding-out at non-eligible institutions, in contrast to the spillover patterns documented by Londoño-Vélez et al. (2020) in Colombia, suggesting that the bulk

of the response comes from genuinely marginal students rather than substitution across institutions. As a result, the composition of incoming cohorts shifted to the left of the ability distribution, a pattern with first-order implications for downstream outcomes.

Estimating effects on educational outcomes is less straightforward, as the change in composition confounds the response of infra-marginal students with that of new entrants. We exploit a feature of the policy design to address this. Cohorts that had already enrolled before 2016 became eligible for *Gratuidad* on their remaining tuition payments, with treatment intensity varying by cohort but with enrollment decisions already made. This holds composition fixed and isolates the response of infra-marginal students. Among these students, the policy reduced dropout by up to 2.5 p.p. and increased on-time graduation by 1.8 p.p. These results suggest a net-positive effect of the two-component policy. For 2016-and-after cohorts, by contrast, we find essentially null effects on dropout, graduation, and persistence, pointing to a negative composition shift that is roughly offset by the policy bundle.

While the reduced-form analysis shows that the policy was successful in expanding access without lowering aggregate completion rates, three reasons motivate a structural approach. First, the difference-in-differences estimates cannot decompose, for each aggregate outcome, the relative contribution of marginal students induced to enroll and infra-marginal students who change their behavior. Second, the policy bundled two changes that the data cannot separately identify without further structure, namely the reduction in out-of-pocket tuition and the elimination of performance requirements. Third, the policy is fiscally costly. Between 2011 and 2019, public spending on higher education in Chile rose by 160% and reached 5.4% of GDP, against 2.9% on average in the rest of the OECD (OECD, 2019). Evaluating budget-improving alternatives that preserve the equity gains of *Gratuidad* requires a model in which students respond to changes in both prices and incentives.

For these reasons, we develop a dynamic discrete-choice model of enrollment, field and institution choice, dropout, and academic effort. In the first stage, a student observes her type (gender, ability, and socioeconomic status) and her personalized choice set (Fack et al., 2019), and decides whether and where to enroll. Conditional on enrollment, in each subsequent period she chooses whether to continue and how much effort to exert, where effort shapes the distribution of credit completion. Performance feeds back into next-period out-of-pocket tuition through the rules of pre-existing scholarship programs, generating a dynamic link between current effort and future financial incentives. Effort is unobserved and recovered from a first-order condition following de Groote (2025), which allows the marginal cost of effort to be estimated jointly with the fixed costs of enrollment. Identification of effort responses relies on the quasi-experimental variation in out-of-pocket tuition induced by the

differential timing of the reform across cohorts, together with the performance-conditional nature of the pre-existing aid system.

We estimate the model using the 2016–2017 entry cohorts, for whom we observe academic progression for up to seven years. Our sequential estimation procedure combines the conditional choice probability approach of Hotz and Miller (1993) with the recovery of effort costs from optimality conditions following de Groote (2025). We account for persistent unobserved heterogeneity using the two-step grouped fixed-effects approach of Bonhomme et al. (2022), classifying students into two latent types based on residual credit-completion patterns. Conditional on these types and observed characteristics, we estimate dropout probabilities and credit-completion distributions, recovering effective effort from the latter. These estimates, together with predicted lifetime earnings, allow us to construct continuation values without repeatedly solving the full dynamic model. We then substitute the effort first-order condition into the value of enrollment, estimate fixed costs by maximum likelihood, and recover marginal effort costs. Because estimation uses post-reform cohorts, the difference-in-differences estimates provide an external benchmark for assessing the model’s simulated responses to the introduction of *Gratuidad*.

Estimation reveals substantial heterogeneity in costs across the student distribution. Fixed costs increase with out-of-pocket tuition, especially at the enrollment stage. They are also higher for female students, for students enrolling in a different region, and for students who have fallen behind in their program. Marginal costs of effort exhibit a clear gradient in both ability and income, with lower-PSU and lower-income students facing higher marginal costs of producing a unit of academic progress. Critically, we find no evidence that the removal of performance requirements weakened effort among infra-marginal students. The composition of new entrants induced by *Gratuidad*, who exhibit higher fixed and marginal costs, accounts for most of the post-reform progression patterns, while the behavior of incumbent students changes little. The model therefore rationalizes the reduced-form evidence. Aggregate completion is held in check by composition rather than by moral hazard.

We use the estimated model to simulate alternative funding designs aimed at reconciling the equity gains of *Gratuidad* with stronger incentives and lower fiscal cost. Three scenarios are central. Re-introducing an annual performance requirement (completing 70% of registered credits) to retain free tuition restores the conditional component removed by the reform and tightens within-cohort incentives. Imposing a minimum PSU score of 510 for eligibility, which harmonizes TFHE with the threshold of pre-existing merit scholarships, reduces the number of beneficiaries while attracting better-prepared students. Combining both elements compounds the selection and incentive effects. The performance requirement raises dropout

among students with high marginal costs but improves progression among those who remain. The PSU requirement is fiscally lighter and improves graduation rates through selection. The joint design preserves a large share of *Gratuidad*'s equity gains while substantially reducing the cost per graduate. We further discuss design variants that target early-career liquidity, differentiate thresholds across academic and vocational programs, or replace hard cutoffs with a smooth eligibility index.

Related literature This paper builds upon several strands of literature. It first contributes to the literature on higher education financing and student outcomes. Broad reviews emphasize that aid design affects not only access, but also persistence, completion, and the timing at which students learn about prices and eligibility (Deming and Dynarski, 2010, Page and Scott-Clayton, 2016, Dynarski et al., 2022). There is substantial evidence showing that increasing grants or reducing net tuition raises enrollment and attainment (Dynarski, 2003, Castleman and Long, 2016, Angrist et al., 2015, Denning, 2017, Fack and Grenet, 2015, Londoño-Vélez et al., 2020, Dobbin et al., 2022, Harris and Mills, 2025). Comparative evidence from Solis, 2017, Molina and Rivadeneyra, 2021, Duryea et al., 2023, and Bolhaar et al., 2024 further shows that the effect of subsidies depends on the pre-reform financing environment and on whether aid takes the form of grants or loans. Evidence on completion and performance is more mixed once composition effects are taken into account (Cohodes and Goodman, 2014, Denning, 2019, Denning et al., 2019, Ferreyra et al., 2024). Beyond partial equilibrium responses, Epple et al., 2006, Epple et al., 2017, and Abbott et al., 2019 show that equilibrium supply responses and redistribution matter for the incidence of education-finance reforms. This paper complements that literature by studying a large-scale tuition-free reform in a setting where the public aid system was already well developed, and by tracing both extensive-margin enrollment responses and post-enrollment outcomes.

It also contributes to the growing literature on tuition-free college and the design of free-college programs. Bucarey (2017) studies the crowding-out effects of Chile's tuition-free reform on low-income students at selective programs. Recent work emphasizes that free-college design is consequential. Unconditional or early-committed aid can generate larger take-up responses than aid that remains uncertain or application-intensive (Burland, 2023, Harris and Mills, 2025), while dynamic models imply that free college often raises enrollment more than graduation because outcomes depend on the type of students induced to enroll and on their effort choices (Matsuda, 2020, Ferreyra et al., 2024). Comparative evidence from promise programs and tuition reversals also documents meaningful heterogeneity across institutional settings (Carruthers and Fox, 2016, Attridge et al., 2025, Bietenbeck et al., 2023). Our contribution is to bring this design question to the Chilean reform and to

quantify how much of the observed effect operates through selection versus changes in effort among infra-marginal students.

Second, we contribute to the literature on the incentive structures and moral hazard inherent in educational funding. Traditional subsidies lower the cost of entry, but they may also weaken the marginal cost of delay or failure. Scott-Clayton (2011) shows that academic incentives embedded in aid rules affect college achievement. Garibaldi et al. (2012) and Beneito et al. (2018) show that continuation tuition and retake pricing can significantly reduce delay and increase effort, while Montalbán (2022) finds that cash support is most effective when accompanied by demanding progress requirements. Related evidence on aid packaging and borrowing frictions (Marx and Turner, 2018, Falco and Reichlin, 2025) suggests that students respond both to the level and to the composition of aid. TFHE also releases students from debt accumulation, which has been shown to affect major choice (Rothstein and Rouse, 2011), dropout (Stinebrickner and Stinebrickner, 2008), career decisions (Sieg and Wang, 2018), and post-college household formation (Black et al., 2023). This paper speaks directly to that mechanism because the Chilean reform simultaneously changed out-of-pocket tuition and removed academic-progress requirements for a large group of students.

Finally, we contribute to a growing structural and semi-structural literature on educational choices under dynamic incentives. A number of papers study responses to changes in incentives by modelling effort or progression as part of a dynamic problem, including Arcidiacono (2005) for affirmative action and admissions, Beffy et al. (2012) for study duration under changing returns to education, and Joensen and Mattana (2024) for grants, loans, and academic progression. Closest in spirit are Ferreyra et al. (2022) in Colombia and Tincani et al. (2023) in Chile, both of which allow incentives and preparedness to shape progression. In a related Chilean setting, Larroucau and Rios (2024) studies retention and reapplication decisions in a centralized admissions market. In our paper, effort is endogenous and recovered from a first-order condition, following de Groote (2025). We contribute by linking current effort to future educational costs through out-of-pocket fees and performance requirements, which allows the model to decompose adverse selection from moral hazard under a large-scale tuition-free reform.

Overview The rest of the paper is organized as follows. Section 2 describes the institutional details, the TFHE policy and introduces the data. Section 3 reports results from differences-in-differences estimation, and Section 4 builds a model of enrollment and effort in higher education. Section 5 discusses the main results of the model and Section 6 performs counterfactual simulations. Section 7 provides a discussion and next steps.

2 Background

This section describes the institutional setting, the policy of interest, and the data used in our analysis. Section 2.1 provides an overview of the Chilean higher education system and the pre-existing funding instruments. Section 2.2 describes the Tuition-Free Higher Education policy introduced in 2016 and the substitution patterns it generated across the previously available aid programs. Section 2.3 introduces the administrative records that we use to track enrollment, performance, and graduation outcomes, and defines the samples used in the reduced-form and structural analyses.

2.1 Institutional Setting

Chilean higher education is well developed, with enrollment rising steadily since the General Law of Universities (*Ley General de Universidades*) of 1981, which incentivized the creation of institutions by allowing entry without state dependence. In 2023, the system comprised a total of 138 institutes: 80 vocational schools (33 Professional Institutes and 47 Centers of Technical Formation) and 58 universities.¹

Universities are divided between the so-called *traditional* universities (a mix of public and private institutions that receive direct support from the government) and *private* universities, which constitute the rest. The latter were created after 1981 and are mainly financed through tuition. Traditional universities are officially known as the Universities of the Rector’s Council (*Consejo de Rectores de las Universidades Chilenas*, *CRUCH*) and are responsible for coordinating the higher education system. They comprise 30 universities and account for around half of total enrollment.²

Access to higher education is through a Centralized Admission Platform, compulsory for traditional universities and some private institutions. Vocational program admission is also possible, although most are conducted off-platform. After high-school completion, students applying to higher education take the centralized admission exam (*Prueba de Selección Universitaria*, PSU). This standardized test, similar to the SAT in the United States, ranges from 150 to 850 points, with a mean of 500 and a standard deviation of 110. Students then submit rank-ordered lists of up to 10 degree programs through the centralized system and are assigned via a deferred-acceptance algorithm (Gale & Shapley, 1962). In practice, the platform imposes a minimum PSU score of 450 to apply.

¹<https://www.ayudamineduc.cl/ficha/instituciones-vigentes-reconocidas-por-el-mineduc>.

²In 2018, the government established a clear set of rules for being considered a traditional university with the promulgation of Law 21091, allowing the first three non-traditional universities to enter.

Higher education is costly in Chile. In 2009, the average tuition fee for a university program was equivalent to 47% of the median family income (Solis, 2017). Costs vary across institution types but remain high even in public institutions. Different funding instruments coexist. Students rely mainly on loans and grants from the Ministry of Education. Eligibility criteria are strongly related to PSU scores. The State Guaranteed Loan program (SGL), introduced in 2006, finances 90% of reference tuition and requires a PSU score of 475 or higher, with no socioeconomic requirement since 2014. Several scholarships also exist, with the *Beca Bicentenario* and *Beca Excelencia* being the most popular. On average, scholarships finance 80% of reference tuition. Eligibility requires a PSU score between 510 and 550, depending on the scholarship, and excludes students in the top two or three income deciles. Both short-cycle programs (SCPs) and university degrees are covered by scholarships. Two-year programs typically require a high-school GPA above 5.0. Access to private loans is limited: in 2015, only 7.5% of student loans came from banks without a state guarantee.

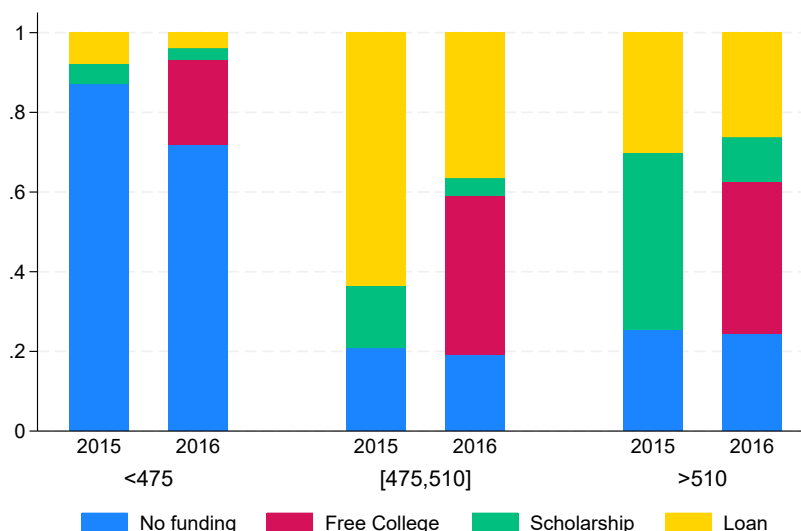
2.2 Tuition-Free Higher Education (TFHE) program

Since the implementation of SGL, student debt rose at a rate of 70% annually, and the number of recipients increased steadily from 15,800 in the first year to 652,000 students in 2016 (Bucarey et al., 2020). In 2011, students led mass protests demanding more affordable higher education. Michelle Bachelet was elected president in 2014 on a platform that included making college free by 2020. In 2015, the TFHE law was enacted, removing tuition fees for students in the bottom half of the income distribution. The policy was first implemented for the 2016 university cohort, expanded in 2017 to include vocational institutes, and further extended in 2018 to the sixth income decile. Applications for TFHE, loans, and scholarships are made through the Centralized Admission Platform by completing a short online form during the application period, regardless of whether the applicant applies to a platform degree or not. A schematic visualization of the policy is provided in [Figure A4](#).

The introduction of TFHE policy generated different substitution patterns depending on students' test scores. [Figure 1](#) shows the distribution of funding instruments before and after the policy, by test score segment. Students scoring below 475 were ineligible for subsidized loans or merit-based scholarships (except for a small scholarship targeting top students from low-SES schools). For this group, TFHE reduced the share of students with no funding. Students scoring between 475 and 510 qualified for subsidized loans but remained below the threshold for most scholarships; for them, substitution was mainly from loans to TFHE, relieving them of any debt upon graduation. Students scoring above 510 qualified for merit-based scholarships (conditional on income eligibility). For this group, substitution was

mainly from scholarships to TFHE, which had two advantages: first, scholarships typically covered only 80% of tuition and fees, whereas TFHE provided full funding; second, scholarships required students to meet satisfactory progression standards (validating about 70% of annual course credits), while TFHE imposed no such requirements. Before the policy, about 20% of scholarship beneficiaries lost eligibility; of these, around 40% switched to subsidized loans, while the rest continued without public funding. After the policy, most substituted to TFHE, while those above the income threshold continued either with loans or no funding, in roughly the same proportions as before.

Figure 1: Distribution of funding instruments, by test score



Notes: This figure shows the distribution of funding instruments by test score group, for 2015 and 2016. PSU is the national standardized test. Students with test scores below 475 are not eligible for public funding, students scoring between 475 and 510 are eligible for loans (CAE and SGL), and students scoring above 510 are eligible for scholarships.

2.3 Data

We use Chilean administrative records that provide detailed student and program data. We observe enrollment, admission test scores, scholarship assignment, socioeconomic information, and demographic characteristics. Institutional data include type, tuition, location, and program length. Between 170,000 and 180,000 students take the national exam each year, of whom about 60% enroll in higher education. Descriptive statistics are reported in [Table B1](#).

In addition, we use restricted data on credit completion from the Information Service of Higher Education (SIES) covering the universe of enrolled students from 2016 to 2023. These

data report the number of courses registered and whether they were completed.

Quasi-Experimental sample We restrict the sample to first-time test takers for the 2012 to 2017 cohort. Since we observe the data until 2023 and the standard university program lasts five years, we allow seven years for students to complete their program. In 2012, scholarships were expanded to include the third income quintile, such that all eligible students for TFHE were already eligible for scholarships conditional on scoring a certain PSU score (Bucarey, 2017).

Model sample We are constrained by the fact that credits are only available from 2016. Together with allowing students 7 years for graduation, we estimate the model on a random sample of 2016-2017 cohorts. CCPs are estimated using the *universe* of 2016-2017 cohorts.

3 Policy Effects

This section provides reduced-form evidence on the effects of the TFHE policy on enrollment and educational outcomes. Section 3.1 presents the differences-in-differences specification and discusses the identifying assumption. In Section 3.2, we estimate the extensive-margin response and document substantial heterogeneity across ability groups and institution types. Section 3.3 addresses the composition problem that arises when estimating intensive-margin effects, and isolates the response of infra-marginal students by exploiting the differential exposure of pre-2016 cohorts to the policy. Section 3.4 discusses the main threats to identification and presents robustness checks.

3.1 Empirical Strategy

We are interested in causally estimating whether the TFHE policy had an impact on enrollment and educational outcomes. To do so, we exploit the exogenous variation in eligibility introduced by the policy and use differences-in-differences. We estimate the following regression model by OLS:

$$Y_{it} = \sum_{\substack{k=2011 \\ k \neq 2015}}^{2020} \beta_k (\mathbb{1}\{t = k\} \times \mathbb{1}\{dec_i < 6\}) + \gamma \mathbb{1}\{dec_i < 6\} + \delta_t + \mathbf{X}_i' \alpha_{\mathbf{x}} + \epsilon_{it} \quad (1)$$

Y_{it} is an outcome of interest including enrollment and educational outcomes such as dropout,

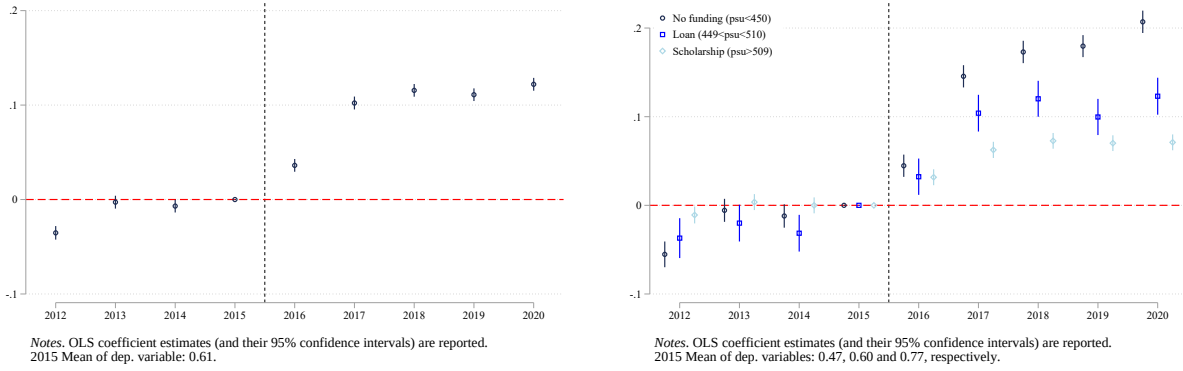
graduation, on-time graduation and persistence, defined as the time (in years) spent in college. $\mathbf{X}_i'$ includes a constant, gender, PSU score, high school GPA, degree type, and mother education. $\mathbb{1}\{dec_i < 6\}$ determines eligibility to TFHE, defined as belonging to the lower half of the income distribution. δ_t are cohort fixed effects. We cluster standard errors at the income decile level using wild bootstrap (Cameron et al., 2008). We use the year before the implementation (2015) as the reference category. β_t is the differences-in-differences estimator.

The identifying assumption is that, in the absence of treatment, eligible and non-eligible students would have experienced similar trends in the outcomes of interest ($\beta_k < 2015$). We test for the validity of this assumption by examining pre-treatment trends. Additionally, we perform several robustness checks, including controlling for differential trends by income decile and restricting the sample to adjacent income deciles. The differences-in-differences estimator can be interpreted as the intent-to-treat effect (ITT) of the policy, given that not all eligible students actually receive TFHE (some do not enroll, while others enroll in non-eligible institutions).

3.2 Effects on enrollment

We first examine the impact of TFHE on the extensive margin, i.e., enrollment. The 2016 cohort is the first for which the policy could affect the enrollment decision. Figure 2 shows the results of estimating Equation 1. As shown in panel a), enrollment among eligible students increased by 3.5 p.p. in 2016 and by around 10 p.p. from 2017 onwards, relative to 2015. This increase in enrollment comes from students for whom the returns to education became positive or who were previously credit constrained. The larger effect, starting in 2017, is consistent with the expansion of free higher education to SCPs. As explained in section 2, the control group are students in the upper half of the income distribution. We then examine heterogeneity by ability groups, which determine loan and scholarship eligibility. In panel b), we observe that lower-ability students, who had limited access to public funding before 2016, are the group that increases enrollment the most. Differences between groups are significant in 2017, when lower-ability students increase enrollment by 14 p.p., compared to 5.5 p.p. for students who already had access to merit scholarships.

Figure 2: Enrollment



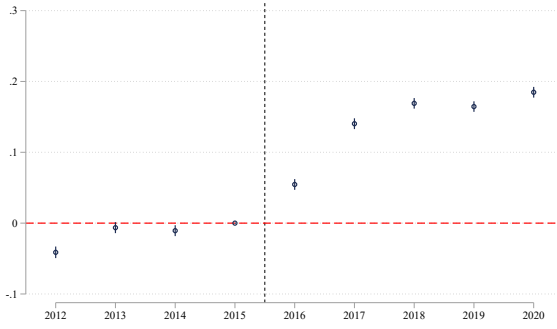
a) Effect on enrollment

b) Effect on enrollment by funding eligibility

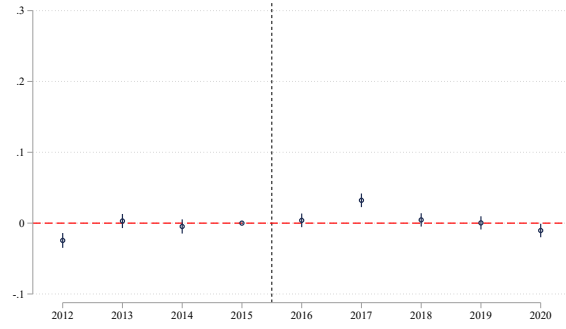
Notes: These figures show the results of estimating Equation 1. Sample is the universe of first time test takers. Panel (a) shows the aggregate results, while panel (b) estimates the equation for groups based on their test scores. PSU is the national standardized test. Students with test scores below 475 are not eligible for public funding, students scoring between 475 and 510 are eligible for loans, and students scoring above 510 are eligible for scholarships.

Substitution between eligible and non-eligible institutions Since TFHE affected around 50% of institutions and 70% of enrollment, it is also important to examine whether there was substitution from non-eligible to eligible centers, or whether enrollment is driven completely by marginal students. Figure 3 reports the estimation on enrollment for eligible and non-eligible institutions. Results show a significant increase of eligible students to higher education institutes that were eligible to the policy. In panel a), we observe increases of 5 and 13 p.p. for years 2016 and 2017, respectively. In contrast to Londoño-Vélez et al. (2020), panel b) suggests no crowding-out. In fact, we find mostly a null effect on enrollment (except for 2017). When looking at panel c), we observe that the increase in enrollment in eligible institutions is driven by low-ability students. Panel d) decomposes the null effect on noneligible institutions in a decrease in enrollment coming from high-ability students while low-ability students increase enrollment around 2 to 4 p.p. A possible explanation is that the policy generated awareness about higher education opportunities and low-ability students that had enough resources opted for higher education even if not eligible to TFHE.

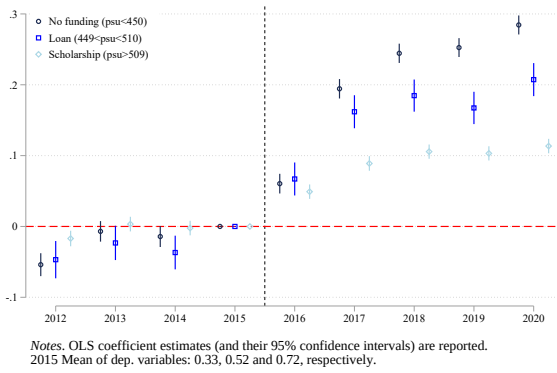
Figure 3: Enrollment by institution eligibility



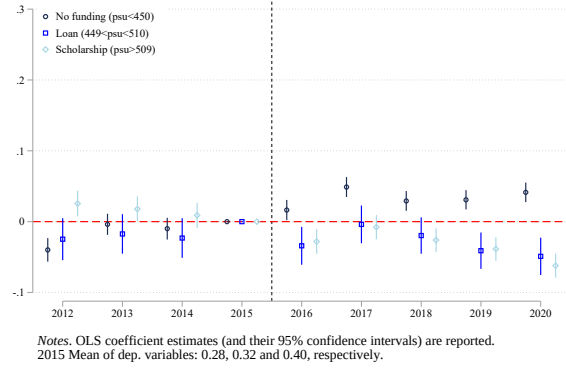
a) Eligible institutions



b) Non-eligible institutions



c) Eligible institutions by funding eligibility



d) Non-eligible institutions by funding eligibility

Notes: These figures show the results of estimating Equation 1 on enrollment for eligible and non-eligible institutions to the TFHE policy. Sample is the universe of first time test takers. Panel a) and b) show the aggregate results, while panels c) and d) disaggregate results by test score group.

Field Preferences We next look at whether students' preferences changed as a result of the policy. We use students' rank-ordered list and construct the dependent variable as the share of a field in student's list. We restrict the sample to eligible institutions and estimate Equation 1 for each field. Results are presented in Figure A5. Panel a) shows that TFHE increased preferences for Health by 1.5p.p. in 2016 and 2p.p. in 2017, while decreasing preferences Arts and Humanities by around 2p.p. STEM majors also lose popularity starting in 2018 with respect to 2015 by 2 p.p. Panel b) decomposes the results into finer categories. The decreases in Arts and Humanities are driven by Education, which decreases by 1p.p., 10% with respect to 2015 average. These results match the evidence found by Castro-Zarzur et al. (2022), which finds that TFHE eclipsed an incentive program to attract talent in

Education majors ³. The increase in Health preferences can be explained by the fact that they are the most expensive programs, along with law majors. Finally, the the 2p.p. decrease in STEM majors is driven by Technology.

3.3 Effects on educational outcomes

Recovering treatment effects for the intensive margin is not as straightforward. If we were to estimate educational outcomes using Equation 1, composition and moral hazard could difficult interpretation: the distribution of students' test scores shifted to the left starting 2016, given that a substantial fraction of lower-ability enrolled (marginal students) as can be seen in Figure A2. On top of that, 2016 cohort onwards eligible students were not subject to performance requirements anymore. To keep the composition of students fixed, we exploit a particularity of the policy: cohorts prior to 2016 were eligible to TFHE for the remaining years of the nominal length of their degrees. For example, if they enrolled in 2015 in a five-year degree, they can be exempted of tuition for the remaining four years, two years for the 2015 cohorts, etc.

We can use this exogenous variation to partial out selection. 2012 to 2015 cohorts are potentially eligible to TFHE starting 2016 but cannot change their initial enrollment decision by construction. We adapt Equation 1 as

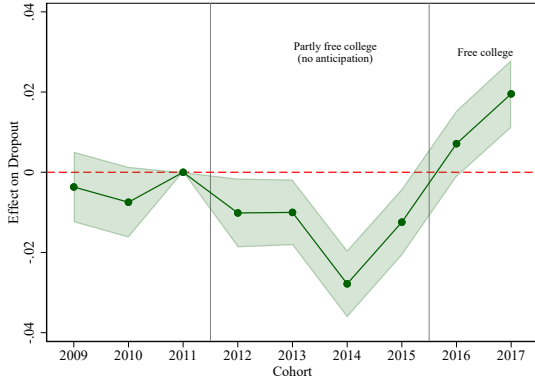
$$Y_{it} = \sum_{\substack{k=2010 \\ k \neq 2011}}^{2017} \beta_k^c (\mathbb{1}\{t = k\} \times \mathbb{1}\{dec_i < 6\}) + \gamma^c \mathbb{1}\{dec_i < 6\} + \delta_t^c + \mathbf{X}_i' \boldsymbol{\alpha}_x^c + \epsilon_{it}^c \quad (2)$$

The above regression remains a standard TWFE differences-in-differences regression, as the time dimension are cohorts. Treatment varies in intensity by cohort, but effects are always estimated with respect to a never treated cohort, so there are no issues of negative weighting (Callaway et al., 2024; de Chaisemartin & D'Haultfœuille, 2024). The variation exploited is purely cross-sectional.

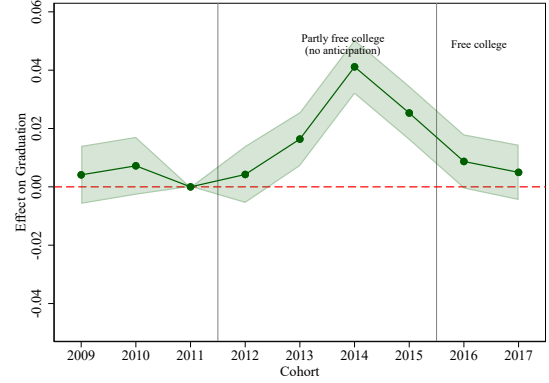
Figure 4 shows the results of estimating Equation 2. Panels a) to d) show the results for dropout, graduation, persistence, and graduation on time, respectively. We find that dropout decrease up to 2.5 p.p. for infra-marginal students, and the effect reverses for 2017 cohort. The composition of new entrants seem to dominate the positive effect from infra-marginal students. Graduation consequently also rises for students that receive the subsidy after their enrollment decision, while cohorts starting 2016 graduate at a similar rate as never-treated students. Persistence results are hard to interpret as the pre-treatment period exhibits a

³see Pal (2025) for a comprehensive analysis of this policy.

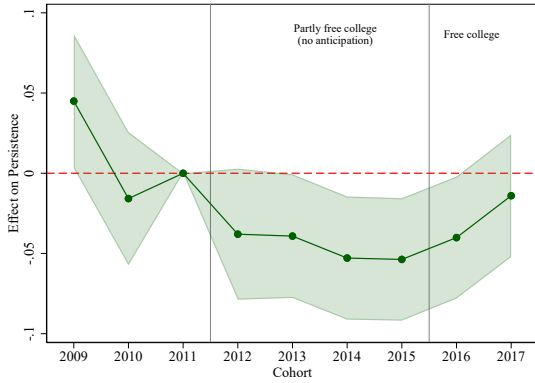
Figure 4: DID outcomes



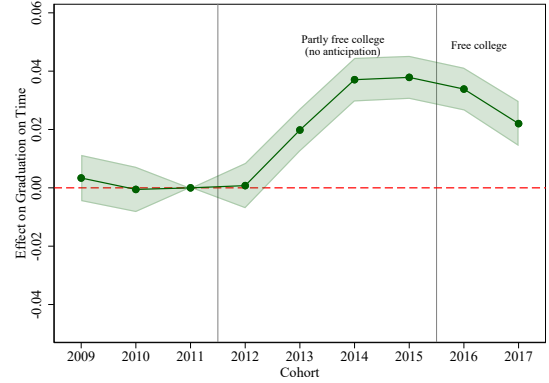
a) Dropout



b) Graduation



c) Persistence



d) Graduation on Time

Notes: These figures show the results of estimating Equation 1 on dropout, graduation, persistence, and graduation on time. Sample is enrolled students. Cohorts 2012 to 2015 are potentially eligible to TFHE starting 2016 but already made their initial enrollment decision. Cohort 2011 is the reference category.

pronounced negative trend, which the effect of policy seems to moderate. Finally, as the subsidy gets larger (2015 cohorts), the proportion of students who graduates on time rises by 1.8 p.p., while there are null effects of TFHE starting 2016.

These results are defined as the overall effect of the policy on those who already enrolled. TFHE lowers the cost of staying enrolled, which tends to reduce dropout and increase graduation. It does so by lowering out-of-pocket and the removal of performance requirements to students who previously benefited from scholarships. Conversely, students with lower PSU scores that now enter higher education face more difficulties to progress and graduate. Quasi-experimental findings suggest that both forces are at play and cancel each other out. The total effect is insignificantly different from zero, meaning the total number of graduates is larger (same graduation rate, but more students). To validate reduced-form results and quantify both effects, we next build a model of enrollment and major choice with endogenous effort.

3.4 Discussion and robustness

Our quasi-experimental analysis suggests that TFHE policy had a positive effect on enrollment on eligible students, especially for low-ability students that previously had limited access to public funding. We find no evidence of crowding-out of non-eligible institutions, suggesting that most of the increase in enrollment comes from marginal students. Eligible institutions were able to absorb the increase in enrollment, especially lower-quality institutions. Regarding educational outcomes, we find modest effects on dropout and graduation for infra-marginal students, while there are no significant effects for students that enrolled after the implementation of the policy, despite being of lower observed ability on average.

There are some limitations to our empirical strategy. We used eligibility to TFHE as a proxy for treatment, and compared eligible to non-eligible students as we do not have a continuous measure of income.

Crowd-out of average ability, high-income students If high-income students were to reduce their enrollment as a result of the policy, our estimates would be biased upwards. However, spillovers are unlikely as cut-offs scores were mostly unchanged, suggesting that the policy did not put additional stress on the capacity of institutions. As a robustness check, we estimate [Equation 1](#) restricting the sample to income deciles 1, 2 and 9, 10, as crowd-out is more likely to occur around the eligibility threshold, and find similar results.

Increase in test-takers If the policy induced more low-income students to take the PSU test, our control group would be positively selected, biasing our results upwards. However, we find no evidence of an increase in test-takers as a result of the policy. This is because take-up rates were already high (around 85% of students in their last year of high school take the test). In fact, what we see is an increase in higher education applications from low income students.

Income manipulation Income is self-reported and verified by the government. Misreporting could lead to students that are actually ineligible being classified as eligible. If this is done on purpose, it could bias our estimates upwards.

4 Model

The decision process of an individual considering attending higher education can be depicted in two stages. In the first stage, she chooses whether to enroll, and if so, which field–institution combination to attend. To decide, she observes program characteristics and available funding instruments, and optimizes how much effort to exert in each program in her personalized choice set, determined by her score in the national exam (Fack et al., 2019). Effort choice will determine her distribution of performance outcomes (credits). Conditional on enrollment, performance today endogenously determines out-of-pocket fees for the subsequent year. The student graduates when she cumulates the required number of credits of the enrolled program.

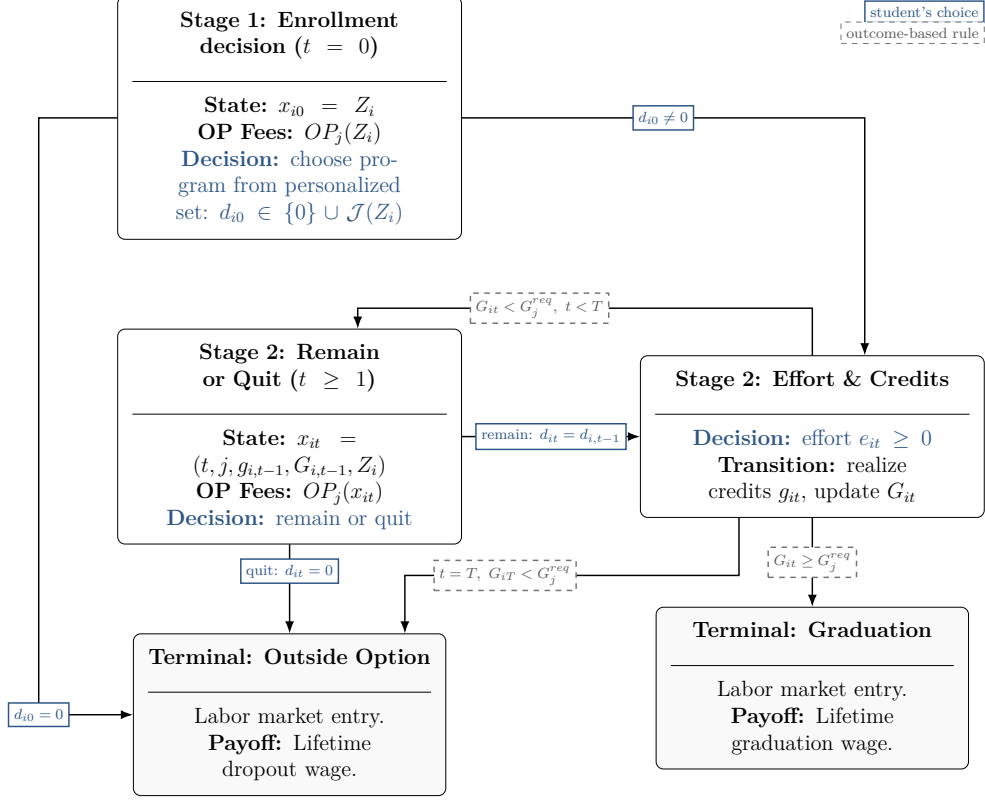
Direct measures of effort are not observed. Instead, we attempt to uncover *effective* effort through realized outcomes, i.e. course credits completion (De Groote, 2025).

4.1 Timing and choices

Figure 5 describes the timing and choices of the model. A student is characterized by her type Z_i , defined by sex, ability (psu and gpa), socioeconomic status and unobserved type. In stage 1, the student decides whether to enroll in higher education. She observes her type Z_i , which characterizes her personalized choice set $\mathcal{J}(Z_i)$ (Fack et al., 2019) and the funding instruments available. She decides between a field–institution combination $d_{i0} = j$, with $j \in \mathcal{J}(Z_i) = \{1, \dots, J(Z_i)\}$ a program from her personalized choice set, and the outside option $j = 0$.

Stage 2 is conditional on enrollment. It is an optimal stopping problem. The student decides in each subsequent period whether to continue in the program, $d_{it} = \{d_{i,t-1}, 0\}$,

Figure 5: Structural model layout



and how much effort $e_{it} \in [0, +\infty)$ to exert. By exerting effort, the student implicitly chooses the probabilities of performance outcomes $g_{i,t+1}$, which accumulates over time as $G_{it} := \sum_{s=1}^t g_{is}$.

The model allows for $T = 7$ academic periods.⁴ They are indexed by $t = 1, \dots, T$. The nominal length of program j is $\bar{t}_j$. A student can choose to drop out before the terminal boundary. In the final academic period she still exerts effort and can graduate; after that performance realization, a nongraduate exits mechanically because an eighth enrollment period is outside the model.

During periods before graduation ($t = 1, \dots, \bar{t}_j - 1$), performance affects both the probability of dropout and, for scholarship holders, out-of-pocket fees $OP_j(x_{it})$: failing to meet the performance requirement ($g_{it} < 3$) implies paying the full tuition the following year. During the graduation period ($t = \bar{t}_j, \dots, T$), the student can graduate and obtain the corresponding wage continuation, postpone graduation when $t < T$, or choose to drop out. In period T , effort and graduation remain endogenous, but a nongraduate is forced to exit after

⁴Since a typical degree program lasts five years, this horizon builds in a two-year margin, allowing completion to be measured with delay.

performance is observed. Hence, the state space is $x_{it} = (t, d_{i,t-1}, g_{i,t-1}, G_{i,t-1}, Z_i)$.

4.2 Utility and the dynamic program

Flow utility from being enrolled in program j during period t includes individual preferences for different programs, out-of-pocket fees sensitivity, and the psychological or physical "disutility" of academic work. We decompose it into a fixed cost (FC_j) that captures preferences for higher education, and a variable cost of effort:

$$u_j(x_{it}) + \varepsilon_{ijt} = -FC_j(x_{it}) - c_j(x_{it})e_{it} + \varepsilon_{ijt}, \quad (3)$$

with marginal cost $c_j(x_{it}) > 0$. Fixed costs $FC_j(x_{it})$ summarize all non-effort related preferences for programs including out-of-pocket fees $OP(Z_i, g_{i,t-1}, P_{j,t})$, defined as the fraction of program fees not covered by the state for student i in program j . These depend on tuition $P_{j,t}$, personal characteristics Z_i and past performance $g_{i,t-1}$:

$$OP(Z_i, g_{i,t-1}, P_{j,t}) = \begin{cases} (1 - \lambda(Z_i, g_{i,t-1}))P_{j,t} & \text{if the student holds a scholarship,} \\ (1 - \lambda(Z_i))P_{j,t} & \text{otherwise.} \end{cases}$$

where $\lambda(\cdot) \in [0, 1]$ is the subsidy rate, i.e. the fraction of program fees $P_{j,t}$ covered by the Government. For scholarship holders, $\lambda(\cdot)$ depends on past performance $g_{i,t-1}$ as well as on individual characteristics Z_i , since failure to meet credit requirements results in the loss of the subsidy. For loans and TFHE, $\lambda(\cdot)$ depends only on observable characteristics and does not vary with performance. Finally, the distribution of the idiosyncratic shock $\varepsilon_{ijt} \sim \text{EV1}$ is iid and common knowledge. It captures what the student learns at the beginning of the period.

The dynamics of the optimization problem are the following: On the one hand, by exerting more effort today, the student i) increases her probability of graduating in subsequent periods and thereby raises her expected future wage, and ii) increases her probability to maintain public funding (passing the performance threshold) if eligible. On the other hand, exerting effort is costly and reduces the flow utility of attending higher education. The conditional value function $v_j(x_{it}, e_{it})$ can be written as

$$v_j(x_{it}, e_{it}) + \varepsilon_{ijt} = u_j(x_{it}, e_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \bar{V}(x_{i,t+1}(\bar{g})) + \varepsilon_{ijt}, \quad (4)$$

where $\bar{V}(x_{i,t+1}(\bar{g}))$ denotes the ex-ante (continuation) value function evaluated at the post-performance state $x_{i,t+1}(\bar{g})$. The first term corresponds to the current flow utility from enrolling in field-institution j . The second term captures the expected continuation value, discounted by β . Uncertainty in $x_{i,t+1}(\bar{g})$ arises solely from the realization of the performance outcome $\bar{g}$.

For a post-performance state, let $\mathcal{A}_{i,t+1}(\bar{g})$ indicate that both staying and dropping out are feasible, and define the active-choice logsum

$$L_{i,t+1}(\bar{g}) = \gamma + \ln \left(\sum_{j' \in \mathcal{J}(x_{i,t+1}(\bar{g}))} \exp(v_{j'}(x_{i,t+1}(\bar{g}))) \right). \quad (5)$$

The continuation value $\bar{V}(x_{i,t+1}(\bar{g}))$ coincides with the logsum $L_{i,t+1}(\bar{g})$ whenever a genuine EV1 choice is available at $t + 1$. Graduation and the terminal boundary, however, are deterministic transitions with no such choice, so the logsum formula does not apply there. We therefore define $\bar{V}$ at the post-performance state piecewise:

$$\bar{V}(x_{i,t+1}(\bar{g})) = \begin{cases} L_{i,t+1}(\bar{g}), & \mathcal{A}_{i,t+1}(\bar{g}) = 1, \\ \alpha_w wage_j(x_{i,t+1}), & \text{graduation,} \\ \alpha_w wage_0(x_{i,t+1}), & \text{terminal nongraduation.} \end{cases} \quad (6)$$

Thus,

$$v_j(x_{it}, e_{it}) + \varepsilon_{ijt} = u_j(x_{it}, e_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} \phi^{\bar{g}}(e_{it}) \bar{V}(x_{i,t+1}(\bar{g})) + \varepsilon_{ijt}.$$

4.3 *Effective* effort and performance

Effective effort e_{it} is unobserved, while performance outcomes g_{it} are directly recovered from students' records. We model performance outcome $g_{i,t+1}$ as the result of effort e_{it} and a logistically distributed shock η_{it} :

$$g_{i,t+1} = \kappa \quad \text{if } \bar{g}_{jt}^\kappa < \ln(e_{it}) + \eta_{i,t+1} \leq \bar{g}_{jt}^{\kappa+1}, \quad (7)$$

where the cutoff values $\bar{g}_{jt}^\kappa$ and $\bar{g}_{jt}^{\kappa+1}$ are field-institution-year specific. In practice, performance is discretized into five categories, $\kappa \in \{0, 1, 2, 3, 4\}$. Students observe the realization of the shock $\eta_{i,t+1}$ only after choosing their effort e_{it} , but since they know the distribution of $\eta_{i,t+1}$ (standard logistic), they can compute the probability of each outcome conditional on effort:

$$\Pr(g_{i,t+1} = \kappa \mid e_{it}, x_{it}) = F(\ln(e_{it}) - \bar{g}_{jt}^\kappa) - F(\ln(e_{it}) - \bar{g}_{jt}^{\kappa+1}), \quad (8)$$

where $F(\cdot)$ denotes the logistic CDF. Making use of the logistic assumption of the performance shock $\eta_{i,t+1}$, for $\kappa = 0$ we can rewrite [Equation 8](#) as:

$$e_{it} = \frac{1 - \Pr(g_{i,t+1} = 0 \mid e_{it}, x_{it})}{\Pr(g_{i,t+1} = 0 \mid e_{it}, x_{it})}. \quad (9)$$

The higher the effort, the higher the expected performance. By choosing effort, students choose the distribution of performance outcomes. *Effective* effort can thus be interpreted as the odds of avoiding the lowest performance outcome (failing all registered credits).

4.4 Solution of the model

The model is solved by backward induction. Higher education is no longer feasible after the seventh performance realization or once the student completes all required program credits.

Formally,

$$\bar{V}(x_{i,T+1}(\bar{g})) = \begin{cases} \alpha_w \text{wage}_j(x_{i,T+1}), & G_{i,T+1} \geq G_j^{\text{req}}, \\ \alpha_w \text{wage}_0(x_{i,T+1}), & G_{i,T+1} < G_j^{\text{req}}. \end{cases} \quad (10)$$

This terminal continuation value is used as an input in earlier periods. Importantly, performance in period T still changes graduation and therefore the relevant wage; only the subsequent enrollment/dropout choice is absent.

At each period t , the student first chooses an effort level e_{it} in her program j . The decision trades off a loss in flow utility with higher future expected gains. Assuming she is rational and behaving optimally, she is solving the following first-order condition for each program ⁵:

$$\frac{\partial v_j(x_{it}, e_{it})}{\partial e_{it}} = \frac{\partial u_j(x_{it}, e_{it})}{\partial e_{it}} + \beta \sum_{g \in \mathcal{G}} \frac{\partial \phi^g(e_{it})}{\partial e_{it}} \bar{V}(x_{it+1}(\bar{g})) = 0 \quad \text{if } e_{it} = e_j^*(x_{it}). \quad (11)$$

At the current-period state x_{it} the choice set $\mathcal{J}(x_{it})$ always contains a genuine choice, so $\bar{V}$ reduces to the plain logsum formula — the same active-choice case used in [Equation 6](#), now evaluated at x_{it} rather than at a post-performance state:

$$\bar{V}(x_{it}) = \gamma + \ln \left(\sum_{j \in \mathcal{J}(x_{it})} \exp(v_j(x_{it}, e_j^*(x_{it}))) \right), \quad (12)$$

and the model is solved recursively by iterating backward to $t = 1$. Given the set of optimal effort levels $\{e_j^*(x_{it})\}_{j \in \mathcal{J}(x_{it})}$, the student chooses the program with the highest value $v_j(x_{it}, e_j^*(x_{it}))$. The resulting choice probabilities take the familiar logit form:

$$p_{jt} = \frac{\exp(v_j(x_{it}, e_j^*(x_{it})))}{\sum_{j' \in \mathcal{J}(x_{it})} \exp(v_{j'}(x_{it}, e_{j'}^*(x_{it})))}.$$

4.5 Identification

As is standard in the dynamic discrete choice literature, the discount factor is not separately identified from the per-period utility function (Magnac & Thesmar, 2002). We therefore fix β at 0.95 and normalize the utility of the outside option to the lifetime expected dropout

⁵a sufficient condition for an interior solution is that students believe there is a non-zero probability to avoid the lowest performance outcome. Positive marginal costs ensure an infinite amount of effort is never exerted.

wage.

We further assume that not enrolling in higher education is an absorbing state: individuals who drop out cannot resit the entrance exam or re-enter higher education in a later period. This restriction rules out option-value effects from delayed re-entry and lets the outside option be summarized by a single terminal payoff. Because this renewal action is available every period, the model exhibits one-period finite dependence: continuation values can be recovered from next-period choice probabilities alone, via the Hotz–Miller inversion (Hotz & Miller, 1993), without full backward induction of the value function (Section 4.6).

4.5.1 Identification of unobserved types

The unobserved component of Z_i is a latent student type, classified once, upstream of the rest of the estimation, and then treated as an additional observed characteristic in every subsequent step. Following the two-step grouped fixed-effects approach of Bonhomme et al. (2022), as implemented empirically by Joensen and Mattana (2024), we classify enrolled students into $M = 2$ latent types via k-means on two individual-average residual moments: completion conditional on registered credits, and completion relative to the intended annual course load. Both moments are residualized on individual and state controls, with period, major, and institution fixed effects, and are standardized before clustering. Types are recovered with k-means++ initialization over 20 independent seeds to guard against local optima, and are labeled from highest to lowest average academic productivity. $M = 2$ is our production classification; $M = 3$ is estimated on the same moments and retained only as an appendix robustness check. Details of the moment construction, the clustering procedure, and a comparison of the $M = 2$ and $M = 3$ classifications are provided in Section Appendix C.

4.5.2 Fixed and Marginal costs

Marginal costs c_j are identified from the FOC (De Groote, 2025). The model attributes higher performance levels, conditional on the same continuation value, to lower marginal costs of effort. Fixed costs then rationalize the observed enrollment shares across programs. The sensitivity to out-of-pocket fees is pinned down by the variation in public funding eligibility, after controlling for observables, which serves as a price shifter.

Wage elasticity α_w is identified through an exclusion restriction. We use local labor market field-region variation. The underlying assumption is that conditional on region and field fixed effects, the interaction provides idiosyncratic local demand shocks that shift utility only through the wage. Specifically, we estimate

$$\begin{aligned} \log(w_j(x_{it})) = & \beta_0^w + \beta_{f,field(j)}^w \text{female}_i + \beta_e^w \text{exper}_{it} + \beta_{e2}^w \text{exper}_{it}^2 \\ & + \sum_{r=1}^{16} \beta_{r,field(j)}^w R_{ir} + \beta_{inst_type(j),field(j)}^w + \varepsilon_{ijt}, \end{aligned} \quad (13)$$

where R_{ir} is a dummy for region r and $inst_type(j) \in \{\text{SCP}, \text{Private}, \text{CRUCH}\}$ denotes the institution type of program j . The gender gap and the institution-type premium are both allowed to vary by field, and the region-field interactions $\{R_{ir} \beta_{r,field(j)}^w\}$ are the source of identifying variation: conditional on region and field fixed effects, they shift utility only through the wage.⁶ Estimation details are provided in Section [Appendix C](#).

4.5.3 Effective effort

We assume students are rational and adjust effective effort so as to maximize expected lifetime utility. As discussed in [subsection 4.3](#), it is a tool to recover the distribution of performance outcomes conditional on state variables. Because effort solves the first-order condition [Equation 11](#) given the state x_{it} , all students sharing the same state x_{it} exert the same level of effort $e_j^*(x_{it})$. As shown in [Equation 7](#), performance outcomes are then generated by an ordered logit model with index $\ln(e_j^*(x_{it}))$, which we can estimate directly as a flexible ordered logit of credit performance, where credits are discretized into five categories (0-4). The exact specification can be found in [Appendix C](#). Performance thresholds are recovered from this estimation, and are allowed to vary by field, institution and year.

4.6 Estimation

We avoid full solution methods by representing the future value term as the conditional value function and a non-negative term that depends on the empirical choice probabilities (Hotz & Miller, 1993). Given the Type-1 extreme value assumption on utility shocks, CCPs are of logit type. We specify flexible parametric functional forms for CCPs, effective effort, fixed costs and wage regression. Details on the estimation of these components are provided in

⁶This equation is estimated on CASEN survey data, which observes broad institution type but not the specific institution attended. To capture within-type variation in earnings, we separately regress log wages on institution and field fixed effects using the Mifuturo administrative dataset, which records graduate earnings by institution but lacks student characteristics. Because part of an institution's Mifuturo fixed effect already reflects its institution type, we demean each institution's fixed effect by subtracting the mean fixed effect across institutions of the same type, so that only the institution's deviation from its type average is added to the CASEN-based wage prediction; this avoids double-counting the type-level premium already identified in [Equation 13](#).

Section [Appendix C](#).

Exiting higher education is a terminal action, so $v_0(x) = \alpha_w \text{wage}_0(x)$. On a branch with a genuine post-performance stay/dropout choice, the Hotz–Miller inversion gives

$$\bar{V}(x_{i,t+1}(\bar{g})) = \gamma + \alpha_w \text{wage}_0(x_{i,t+1}) - \ln p_{0,t+1}(x_{i,t+1}(\bar{g})). \quad (14)$$

This expression is not applied mechanically to graduation or to the hard terminal boundary. Combining [Equation 14](#) with [Equation 6](#),

$$\bar{V}(x_{i,t+1}(\bar{g})) = \begin{cases} \gamma + \alpha_w \text{wage}_0(x_{i,t+1}) - \ln p_{0,t+1}(x_{i,t+1}(\bar{g})), & \mathcal{A}_{i,t+1}(\bar{g}) = 1, \\ \alpha_w \text{wage}_j(x_{i,t+1}), & \text{graduation,} \\ \alpha_w \text{wage}_0(x_{i,t+1}), & \text{terminal nongraduation.} \end{cases} \quad (15)$$

Accordingly, the dropout logit is estimated only on behavioral risk-set observations before the terminal period: period $T - 1$ predicts the decision to return for the final period T .

From the first-order condition in [Equation 11](#), we can directly recover the expression for marginal costs:

$$c_j(x_{it}) = \beta \sum_{\bar{g} \in \mathcal{G}} \frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}} \bar{V}(x_{it+1}(\bar{g})) \quad \text{if } e_{it} = e_j^*(x_{it}), \quad (16)$$

where $\bar{V}(x_{it+1}(\bar{g}))$ is computed using the estimated CCPs and dropout wage, and $\frac{\partial \phi^{\bar{g}}(e_{it})}{\partial e_{it}}$ can be directly computed given the logistic assumption about η_{t+1} in [Equation 7](#).

Replacing the expression for marginal costs in the flow utility, and defining

$$a_{ijt}^{\bar{g}} = \phi^{\bar{g}}(e_{ijt}^*) - e_{ijt}^* \frac{\partial \phi^{\bar{g}}(e_{ijt})}{\partial e_{ijt}} \Big|_{e_{ijt}=e_{ijt}^*}. \quad (17)$$

The continuation component is split as

$$EMAX_{ijt} = \sum_{\bar{g} \in \mathcal{G}} a_{ijt}^{\bar{g}} \mathbb{1}\{\mathcal{A}_{i,t+1}(\bar{g}) = 1\} [\gamma - \ln p_{0,t+1}(x_{i,t+1}(\bar{g}))], \quad (18)$$

$$EWAGE_{ijt} = \beta \sum_{\bar{g} \in \mathcal{G}} a_{ijt}^{\bar{g}} wage_{\bar{g}}(x_{i,t+1}), \quad (19)$$

$$v_j(x_{it}, e_{ijt}^*) = -FC_j(x_{it}) + \beta EMAX_{ijt} + \alpha_w EWAGE_{ijt}. \quad (20)$$

Here $wage_{\bar{g}}$ equals the graduation wage on graduation branches and the dropout wage otherwise. At the terminal period T , $EMAX_{ijt} = 0$ by construction, while $EWAGE_{ijt}$ remains nonzero and preserves the effort incentive operating through the probability of graduation.

Estimation can be performed following four sequential steps.

1. We first classify students into unobserved types via k-means, as described in Section 4.5.1. This classification is computed once and thereafter treated as an additional element of Z_i in every subsequent step.
2. We flexibly estimate $p_{0,t+1}$ from a logistic regression and $\ln e_j^*(x_{it})$ the underlying index of an ordered logit model of credit completion (0-4).
3. The FOC equates marginal cost to marginal benefits, with an expression that can be directly estimated from the data. We can substitute the expression in $v_j(x_{it})$ and estimate $FC_j(x_{it})$ by maximum likelihood.
4. Finally, we use the estimates and the FOC to compute $c_j(x_{it})$.

5 Results

This section first presents reduced-form estimates that predict dropout probabilities and performance outcomes. Then, structural parameters are discussed, which include students fixed and marginal costs.

5.1 Dropout probabilities

Table 1 reports the results of a logistic regression model of dropout. Overall, dropout probabilities depend strongly on the enrollment length. Progressing through the program reduces the likelihood of dropping out (relative to period 1). Being female reduces the

Table 1: Dropout probabilities

	dropout	
OP fees	0.101***	(0.006)
Rel. Ability	-0.105***	(0.023)
Diff. Region	0.106***	(0.026)
Credits left	-0.113***	(0.004)
Designed total credits	0.099***	(0.006)
Delay	1.098***	(0.051)
Performance: category 0	2.116***	(0.040)
Performance: category 1	1.435***	(0.040)
Performance: category 2	0.291***	(0.033)
Performance: category 4	-0.721***	(0.049)
female	0.197**	(0.079)
Middle SES	0.033	(0.098)
High SES	0.007	(0.100)
PSU	0.063	(0.066)
GPA	-0.111**	(0.053)
Type 2	-10.276***	(0.656)
OP $\times$ female	-0.010**	(0.005)
OP $\times$ middle SES	-0.030***	(0.007)
OP $\times$ high SES	-0.027***	(0.008)
OP $\times$ psu	-0.026***	(0.004)
OP $\times$ gpa	0.015***	(0.003)
Delay $\times$ female	0.057	(0.050)
Delay $\times$ middle SES	0.040	(0.062)
Delay $\times$ high SES	0.233***	(0.062)
Delay $\times$ PSU	-0.191***	(0.035)
Delay $\times$ GPA	-0.062*	(0.034)
Delay $\times$ SCP	0.584***	(0.042)
t=2 $\times$ SCP	0.446***	(0.030)
t=3 $\times$ SCP	0.832***	(0.047)
t=4 $\times$ SCP	0.988***	(0.087)
t=5 $\times$ SCP	0.830***	(0.184)
t=6 $\times$ SCP	0.910**	(0.391)
Performance 0 $\times$ female	-0.301***	(0.039)
Performance 0 $\times$ middle SES	-0.060	(0.046)
Performance 0 $\times$ high SES	-0.093*	(0.051)
Performance 0 $\times$ PSU	-0.022	(0.027)
Performance 0 $\times$ GPA	-0.011	(0.026)
Performance 1 $\times$ female	-0.278***	(0.041)
Performance 1 $\times$ middle SES	-0.024	(0.048)
Performance 1 $\times$ high SES	0.053	(0.053)
Performance 1 $\times$ PSU	0.005	(0.027)
Performance 1 $\times$ GPA	0.072***	(0.027)
Performance 2 $\times$ female	-0.099***	(0.035)
Performance 2 $\times$ middle SES	-0.028	(0.043)
Performance 2 $\times$ high SES	0.082*	(0.046)
Performance 2 $\times$ PSU	-0.103***	(0.024)
Performance 2 $\times$ GPA	0.079***	(0.024)
Performance 4 $\times$ female	-0.328***	(0.052)
Performance 4 $\times$ middle SES	-0.046	(0.065)
Performance 4 $\times$ high SES	0.263***	(0.065)
Performance 4 $\times$ PSU	-0.180***	(0.036)
Performance 4 $\times$ GPA	-0.091**	(0.035)
Credits left $\times$ female	0.001	(0.002)
Credits left $\times$ middle SES	0.001	(0.003)
Credits left $\times$ high SES	-0.007**	(0.003)
Credits left $\times$ psu	0.007***	(0.002)
Credits left $\times$ gpa	0.010***	(0.002)
Rel. Abil. $\times$ female	0.038***	(0.011)
Rel. Abil. $\times$ middle SES	0.011	(0.013)
Rel. Abil. $\times$ high SES	0.057***	(0.015)
Rel. Abil. $\times$ psu	0.011*	(0.006)

probability of dropping out, with the effect becoming stronger as students progress in their studies.

Observed ability, measured as GPA during high school and exam entry score (PSU), is strongly correlated with persistence in higher education, but conditional on the student's progress (accumulated credits), its estimated coefficient diminishes. Both cumulative credits and the number of credits completed in the previous year are strong determinants of student progression. Being delayed in the program increases the probability of dropping out, with a stronger magnitude for females.

Type 2 capture students that are very sensitive to performance, and have a higher probability of dropping out when being delayed and out-of-pocket fees are high.

5.2 Performance

[Table B3](#) presents the results of the ordered logit model of performance with five categories (0–4). Coefficients follow a similar pattern as in the logistic dropout model. Ability is a strong predictor of performance, as is income. Out-of-pocket fees have a negative effect on performance, which is stronger for low-income and low-ability students.

Being delayed in the program reduces the probability of obtaining credits, and the effect is stronger for females. Finally, past credits is a strong predictor of current performance, particularly for low-ability students.

Type 2 students are worse performers, but they are hurt less than type 1 from being delayed and from out-of-pocket fees.

5.3 Fixed Costs

Estimates of fixed costs and expected wage are presented in [Table 2](#). Students' fixed costs increase with out-of-pocket fees, particularly at the program choice stage. Once enrolled, this effect is substantially attenuated in the continuation decision. High SES students are less sensitive to out-of-pocket fees than their low SES counterparts, while the opposite is true for high-ability students. Students also face higher fixed costs when the program is located in a different region.

Past progression significantly affects fixed costs, as reflected in the coefficient of the delay variable. Again, females face higher fixed costs than males when delayed, and students who performed well on the national exam appear to experience smaller penalties from lagging

Table 2: Fixed costs

<i>Fixed Costs</i>		
Enrol HE	2.959***	(0.144)
Enrol HE $\times$ Elite	-0.815***	(0.098)
Enrol HE $\times$ Other CRUCH	0.276***	(0.081)
Enrol HE $\times$ G8	0.370***	(0.122)
OP fees	0.024***	(0.001)
OP $\times$ low SES	0.025*	(0.013)
OP $\times$ middle SES	-0.002	(0.013)
Diff. Region	0.935***	(0.061)
Diff. Region $\times$ female	-0.178***	(0.051)
Diff. Region $\times$ low SES	0.417***	(0.068)
Diff. Region $\times$ middle SES	0.414***	(0.072)
Diff. Region $\times$ psu	0.197***	(0.034)
Diff. Region $\times$ gpa	0.221***	(0.034)
Relative ability	0.494***	(0.062)
Rel. Abil. $\times$ female	0.066***	(0.022)
Rel. Abil. $\times$ low SES	-0.426***	(0.028)
Rel. Abil. $\times$ middle SES	-0.265***	(0.030)
Performance (t-1): category 0	2.670***	(0.130)
Performance (t-1): category 1-2	-0.064	(0.092)
Performance (t-1): category 4	-0.968***	(0.087)
Credits (t-1) $\times$ female	-0.179***	(0.024)
Credits (t-1) $\times$ low SES	-0.024	(0.032)
Credits (t-1) $\times$ middle SES	0.005	(0.034)
Credits (t-1) $\times$ psu	-0.042**	(0.017)
Credits (t-1) $\times$ gpa	0.018	(0.016)
Credits left (t-1)	0.090***	(0.011)
Credits left (t-1) $\times$ female	0.018**	(0.008)
Credits left (t-1) $\times$ low SES	-0.048***	(0.011)
Credits left (t-1) $\times$ middle SES	-0.039***	(0.011)
Credits left (t-1) $\times$ psu	-0.052***	(0.006)
Credits left (t-1) $\times$ gpa	-0.005	(0.005)
Type 2	-1.067***	(0.044)
EMax	0.950	(.)
Expected wage (USD million)	23.826***	(1.139)
Expected wage $\times$ low SES	2.820**	(1.268)
Expected wage $\times$ middle SES	2.935**	(1.383)
Product FE	Yes	

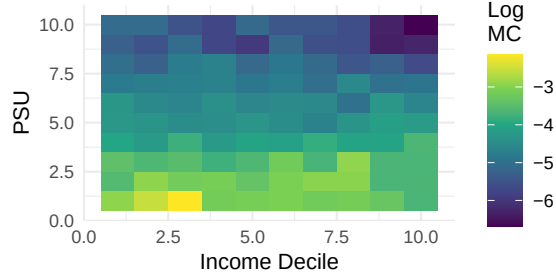
Standard errors in parentheses

Fixed Costs and expected wage

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

behind. Expected wage has a positive effect on the conditional value function, indicating that students value their future expected earnings.

Figure 6: Marginal Costs



Notes: Heatmap of log marginal costs by PSU (y-axis) and income (x-axis) decile.

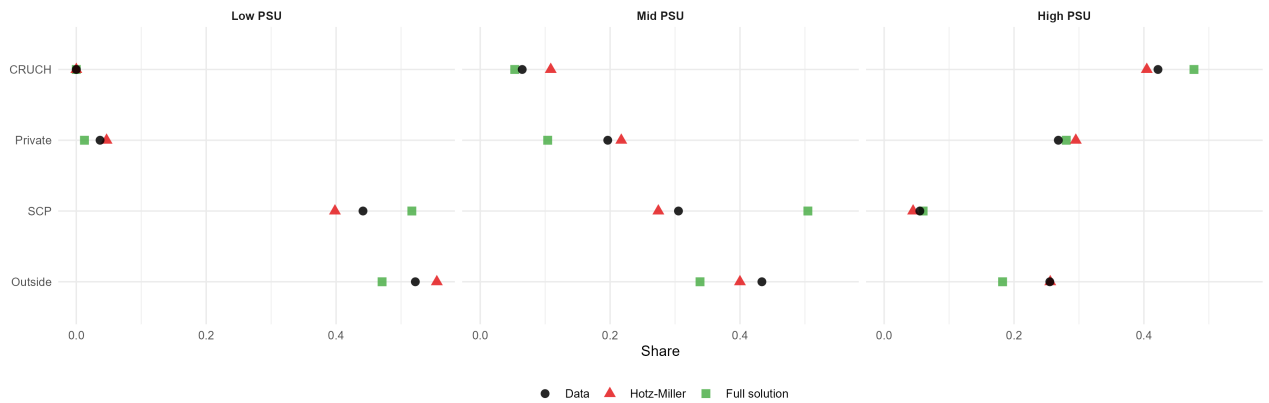
Figure 8 shows the distribution of log marginal costs by income and PSU score. The heatmap exhibits a gradient in marginal costs in both ability and socioeconomic status dimensions.

Table 3 presents, for interpretational purposes, the unit-weighted projection of log marginal costs over the complete reachable state space. The fitted marginal cost decreases with PSU score and high-school GPA and increases with out-of-pocket fees and prior delay. The interaction coefficients allow these gradients to vary with student characteristics.

5.4 Model Fit

5.4.1 In-sample Validation

Figure 7: Fit by ability tercile



Notes: In sample fit by PSU tercile. Dynamic model with endogenous effort.

Table 3: Marginal costs

	log(mc)	
female	-0.408***	(0.062)
Low SES	-0.372***	(0.079)
Middle SES	-0.605***	(0.084)
psu	-0.306***	(0.060)
gpa	-0.214***	(0.046)
Type 2	3.314***	(0.016)
OP fees	-0.019***	(0.006)
Diff. Region	0.220***	(0.024)
Credits completed (t-1)	-0.781***	(0.013)
Credits left (t-1)	0.032***	(0.005)
t = 2	3.129***	(0.039)
t = 3	3.459***	(0.038)
t = 4	2.998***	(0.041)
t = 5	2.426***	(0.050)
t = 6	3.423***	(0.056)
t = 7	4.067***	(0.154)
Credits (t-1) $\times$ female	-0.017**	(0.007)
Credits (t-1) $\times$ low SES	0.023***	(0.009)
Credits (t-1) $\times$ middle SES	0.048***	(0.010)
Credits (t-1) $\times$ psu	0.063***	(0.006)
Credits (t-1) $\times$ gpa	-0.026***	(0.005)
Credits left (t-1) $\times$ female	0.000	(0.003)
Credits left (t-1) $\times$ low SES	0.017***	(0.004)
Credits left (t-1) $\times$ middle SES	0.017***	(0.004)
Credits left (t-1) $\times$ psu	-0.008***	(0.003)
Credits left (t-1) $\times$ gpa	-0.010***	(0.002)
OP $\times$ female	0.015***	(0.004)
OP $\times$ low SES	0.040***	(0.007)
OP $\times$ middle SES	0.025***	(0.006)
OP $\times$ psu	-0.009***	(0.003)
OP $\times$ gpa	-0.011***	(0.003)
Diff. Region $\times$ female	-0.008	(0.019)
Diff. Region $\times$ low SES	-0.125***	(0.025)
Diff. Region $\times$ middle SES	-0.097***	(0.027)
Diff. Region $\times$ psu	-0.016	(0.014)
Diff. Region $\times$ gpa	-0.020	(0.014)
Observations	19,606	
Field FE	Yes	
Institution FE	Yes	

Standard errors in parentheses

MC heterogeneity.

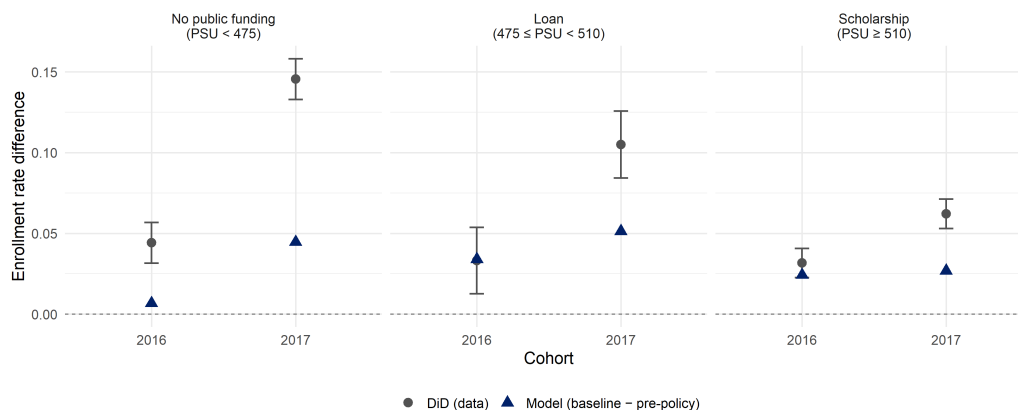
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5.4.2 Out-of-sample Validation

The structural model is estimated entirely on post-reform (2016–2017, *Gratuidad*-era) data, so its parameters are never directly disciplined by pre-reform behavior. To check that the model is not simply curve-fitting the post-reform scenario, we use it to simulate a *pre-policy* counterfactual that reverts the funding rule to the regime that prevailed before 2016: the *Gratuidad* coverage rate is set to zero, so eligible students face the same PSU/income-conditional loan and performance-contingent merit-scholarship schedule as everyone else, with no unconditional tuition benefit.

We then compare the model-implied difference between the *Gratuidad* baseline and this simulated pre-policy regime to the reduced-form difference-in-differences estimates of [subsection 3.2](#) and [subsection 3.3](#), separately for (i) overall enrollment and enrollment by PSU-eligibility tercile (matching the DiD instrument groups: no public funding, loan-only, and scholarship-eligible), and (ii) intensive-margin outcomes — graduation and dropout — among the cohorts affected by the DiD design. Because the DiD variation is never used to fit the structural parameters, agreement between the two designs is informative about whether the model captures the reform’s actual behavioral response rather than an artifact of the post-reform sample it was estimated on.

Figure 8: Model vs. DiD: Enrollment



Notes:

The model can replicate 2016 effects except for low ability students, but fails short for the 2017 cohort enrollment jump. Predicted enrollment increase in 2017 is around 10 p.p. while the model predicts 5 p.p.

6 Counterfactuals

We use the estimated model to simulate counterfactual funding policies, evaluating impacts on both enrollment and intensive (effort, credit completion, time-to-degree, graduation) margins. Each policy is defined as a change to the funding rule $OP(x_{it})$ (income and PSU eligibility cutoffs, coverage rate, and performance conditioning), holding the estimated cost primitives $FC_j(\cdot)$, $c_j(\cdot)$, and wages fixed. For each scenario we resolve the full backward induction over the state space and forward-simulate outcomes by Monte Carlo, drawing performance and choice shocks many times per student to average out simulation noise, with effort chosen optimally each period from the FOC in Equation 11. Throughout, the reference scenario is the estimated pre-TFHE regime; each counterfactual is reported as a change relative to this reference.

6.1 Decomposing the effects of each policy

For each counterfactual we decompose the change in outcomes into three channels that are mutually exclusive and exhaustive: a *selection* effect from students whose enrollment decision changes, and, among students who would enroll under either regime, a *price* effect and an *effort* effect.

Let $r \in \{1, \dots, R\}$ denote the policy regime and $D_i^r \in \{0, 1\}$ the enrollment decision of student i under regime r . Define the infra-marginal cohort as the students who enroll under the reference (status-quo) regime r' ,

$$\mathcal{IM} = \{i : D_i^{r'} = 1\},$$

fixed across all counterfactuals so that changes computed on $\mathcal{IM}$ cannot reflect a change in who is enrolled — a policy that excludes some of these students (e.g. through a PSU cutoff) counts them as non-graduates rather than shrinking the denominator. Let Y_{it}^r denote a potential outcome (e.g. graduation) under regime r , written in the structural model as $Y_{it}^r = Y_{it}(FC_j^r(x_{it}), c_j^r(x_{it}), e_j^r(x_{it}))$. The total effect on the full population decomposes as

$$\underbrace{\mathbb{E}[Y_i^r] - \mathbb{E}[Y_i^{r'}]}_{\Delta Y_{\text{total}}} = \underbrace{\Delta Y_{\text{total}} - \Delta Y_{\text{behav}}}_{\Delta Y_{\text{selection}}} + \underbrace{\Delta Y_{\text{price}}}_{\text{effort fixed}} + \underbrace{\Delta Y_{\text{behav}} - \Delta Y_{\text{price}}}_{\Delta Y_{\text{effort}}},$$

where $\Delta Y_{\text{behav}} = \mathbb{E}[Y_i^r - Y_i^{r'} \mid i \in \mathcal{IM}]$ is the change among infra-marginal students with effort re-optimized, and ΔY_{price} is the same quantity computed with $e_j^r(x_{it})$ mechanically fixed at its pre-policy (status-quo) value, so that any remaining change for $\mathcal{IM}$ can only

come from the reduction in out-of-pocket cost. $\Delta Y_{\text{selection}}$ is therefore identified as the residual composition effect of marginal entrants, and ΔY_{effort} isolates the part of the behavioral response that operates through re-optimized studying intensity rather than through relaxed financial constraints — a channel that only the endogenous-effort model can separate from the price channel, since a reduced-form design can at best separate $\Delta Y_{\text{selection}}$ from ΔY_{behav} . We apply this decomposition to each of the scenarios below, and additionally report outcomes split by PSU tercile, SES/income group, and by the latent unobserved type $\tau_i \in \{1, 2\}$ recovered in [subsubsection 4.5.1](#).

6.2 PSU and performance requirements

We saw that TFHE substitutes most of the preexisting aid programs, which were conditional on performance. The removal of performance requirements may have weakened incentives for effort and progression. Preexisting aid programs had performed well in attracting students (Falco & Reichlin, 2025; Solis, 2017, 2026), but the out-of-pocket cost could still be over 20% of tuition. From the Government’s perspective, the current policy is very costly: between 2011 and 2019, spending on higher education increased by 160%, reaching 5.4% of GDP, compared with 2.9% in the rest of the OECD (OECD, 2019).

The counterfactuals below build on the conditions of the preexisting aid programs described in [subsection 2.1](#): subsidized loans required $PSU \geq 475$, and merit scholarships required $PSU \geq 510$ and completion of at least 70% of registered credits each year (three of four credit-units in the model’s discretized performance measure, $g_{t-1} \geq 3$). They are designed to be budget-improving while preserving the equity target of TFHE. We report four post-reference scenarios: the unrestricted *Gratuidad* reform and three designs that modify the funding rule $OP(x_{it})$ applying specifically to TFHE. The pre-existing loan and scholarship schedules are left untouched in all four scenarios. The three alternative designs are:

1. **Performance requirement.** Impose the same annual performance requirement used by the pre-existing scholarships ($g_{t-1} \geq 3$) to *retain* free tuition. This restores the performance-contingent component removed by TFHE, without changing who is initially eligible.

Model prediction: first-period fixed and marginal costs are unaffected, since eligibility is unchanged. From the second year on, students at risk of falling below the threshold face a discrete jump in $OP(x_{it})$ if they fail to comply, so those with the highest marginal and fixed costs of effort should raise effort the most (or drop out if the increase is too costly). If the expected utility of a first year is not compensated by this option value, some students may not enroll at all.

2. **PSU requirement.** Impose $PSU \geq 510$ for *Gratuidad* eligibility, harmonizing the threshold with the pre-existing merit scholarship. This changes the extensive margin only: no performance condition is added, and no fixed or marginal cost changes for students who remain eligible.

Model prediction: mechanically reduces Government expenditure by shrinking the eligible pool, and attracts, on net, higher-ability students with lower fixed and marginal costs, so persistence and graduation among the enrolled population should improve through selection alone.

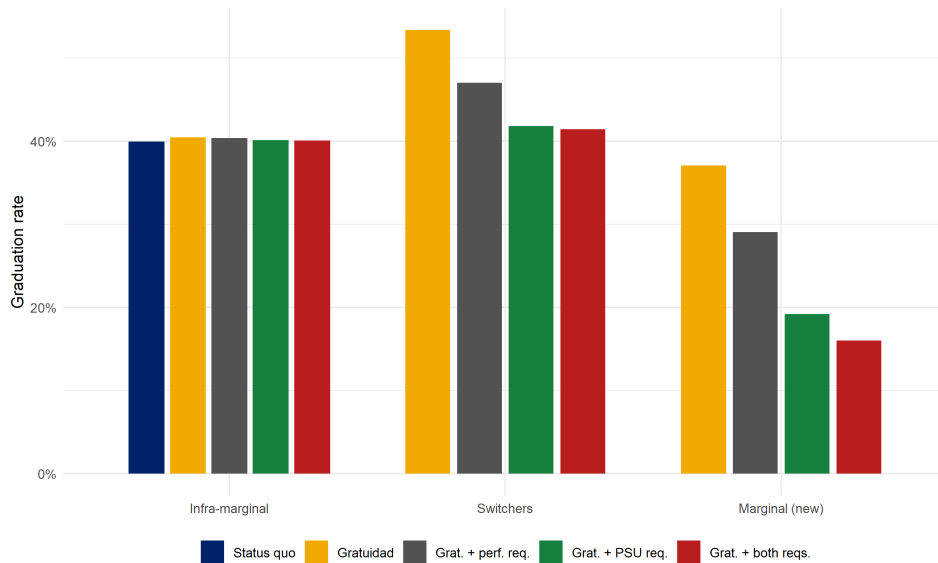
3. **Joint requirement.** Combine (1) and (2): $PSU \geq 510$ for eligibility, and the performance requirement to retain funding once enrolled.

Model prediction: combines the selection and incentive channels above. Reduces the number of beneficiaries the most of the three scenarios, but the remaining recipients face both a more favorable cost distribution and stronger incentives to progress.

Mechanically, (1) affects $OP(x_{it})$ via g_{t-1} for *all* TFHE recipients, tightening incentives within a cohort; (2) affects selection into TFHE at $t = 0$; and (3) compounds both.

Table 4 summarizes the counterfactual results. The three panels distinguish enrollment, graduation, and graduation conditional on enrollment. Entries are percentage-point changes relative to the simulated pre-policy regime, so the columns can be compared directly.

Figure 9: Graduation rate by margin group



Notes: Simulated graduation rate by margin group (infra-marginal students enrolled under both regimes, switchers, and new marginal entrants) and counterfactual scenario, as defined in [subsection 6.1](#).

Table 4: Counterfactual effects on enrollment, graduation and expected earnings

	Status quo level	Change relative to status quo			
		Gratuidad	Before-HE performance requirement	During-HE performance requirement	Both requirements
<i>A. Enrollment (% , p.p.)</i>					
All students	57.0	+2.72	+1.12	+2.10	+0.94
Inframarginal students	100.0	0.00	0.00	0.00	0.00
Income-eligible students	50.0	+4.00	+1.64	+3.08	+1.34
Low ability	42.1	+2.64	0.00	+1.78	0.00
Average ability	50.6	+3.46	+1.31	+2.80	+1.09
High ability	78.5	+2.05	+2.05	+1.72	+1.72
<i>B. Overall graduation (% , p.p.)</i>					
All students	22.8	+1.55	+0.64	+1.15	+0.53
Inframarginal students	40.0	+0.51	+0.16	+0.42	+0.13
Income-eligible students	17.3	+2.35	+0.93	+1.70	+0.77
Low ability	10.1	+1.54	0.00	+0.91	0.00
Average ability	19.3	+1.88	+0.70	+1.55	+0.59
High ability	39.1	+1.22	+1.22	+1.00	+1.00
<i>C. Graduation conditional on enrollment (% , p.p.)</i>					
All students	40.0	+0.77	+0.33	+0.53	+0.27
Inframarginal students	40.0	+0.51	+0.16	+0.42	+0.13
Income-eligible students	34.7	+1.78	+0.71	+1.19	+0.59
Low ability	24.1	+2.02	0.00	+1.10	0.00
Average ability	38.1	+1.04	+0.39	+0.91	+0.34
High ability	49.7	+0.25	+0.25	+0.18	+0.18
<i>D. Expected lifetime earnings (thousand 2015 USD, %)</i>					
All students	129.4	+2.42	+1.08	+1.87	+0.89
Inframarginal students	164.7	+0.35	+0.18	+0.32	+0.18
Income-eligible students	115.6	+4.30	+1.87	+3.29	+1.54
Low ability	100.4	+2.90	0.00	+1.98	0.00
Average ability	119.8	+3.26	+1.33	+2.58	+1.09
High ability	168.3	+1.55	+1.55	+1.29	+1.29

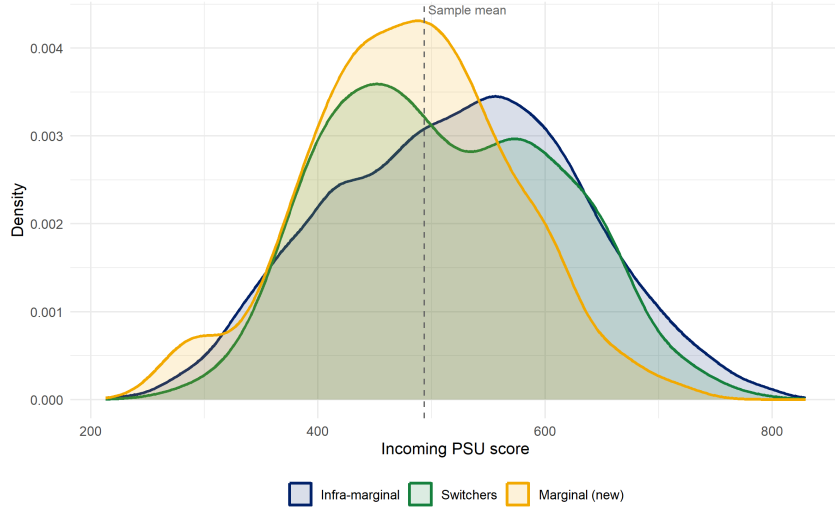
Notes: The first column reports the average level of each outcome for each group under the simulated pre-policy regime (status quo). The remaining columns are changes relative to that level. The units in parentheses in each panel title are those of the status-quo level and of the change: percentage points (p.p.) for enrollment and graduation, percent of the status-quo level for earnings. Gratuidad is unrestricted; the other columns add the named condition. The before-HE performance requirement is the PSU cutoff required to obtain Gratuidad at entry; the during-HE performance requirement is the academic-progress condition for keeping Gratuidad.

Overall graduation uses the full group and counts non-enrollees as non-graduates. Conditional graduation uses the enrollees in each scenario.

Expected lifetime earnings are the present value (5% discount rate, 40 years of experience) of model-predicted earnings, averaged over all students in the group. Non-enrollees earn the no-higher-education wage, graduates the degree wage of their programme, and dropouts the dropout wage, which rises with credits earned. Predictions come from the wage regressions without shocks and measure earnings only: they exclude tuition, effort and other costs, so they are not a welfare measure.

Inframarginal students are those who enroll in the same programme both under the simulated pre-policy regime and under Gratuidad (the same programme, given common random numbers across scenarios); membership is fixed across scenarios, so the change in enrollment in this row is the fraction of them who stop enrolling under a restricted scenario. Students who enroll under both regimes but in a different programme are excluded. Ability is proxied by PSU terciles. Nearly all low-ability students are excluded from Gratuidad eligibility under the before-HE performance requirement, so their near-zero deltas there reflect that exclusion, not a lack of data.

Figure 10: Incoming ability (PSU) by margin group



Notes: Distribution of incoming PSU score by margin group. The dashed line marks the sample mean.

7 Conclusion

This paper investigated the impacts of Tuition-Free Higher Education policies on student enrollment, persistence, and academic performance, using the 2016 Chilean reform as a natural experiment. Our difference-in-differences analysis reveals that the policy significantly boosted enrollment amongst eligible students by 10p.p., driven by lower ability students. Effects on educational outcomes were estimated for infra-marginal students, with a decrease in dropout of 2 p.p. and an increase in on-time graduation of 1.8 p.p.

We built a structural model that endogenizes effort choices, capable of disentangling selection effects from behavioral responses, finding that the removal of performance-based requirements did not lead to reduced effort or weakened outcomes overall. The policy induced a composition effect by attracting student with a higher cost of effective effort, while infra-marginal students responded to the policy by increasing effort, as their marginal cost of effort diminished with the reduction in out-of-pocket costs.

These findings have important implications for higher education financing worldwide, suggesting that unconditional aid can promote equity and mobility, especially in contexts with high tuition costs and imperfect income verification.

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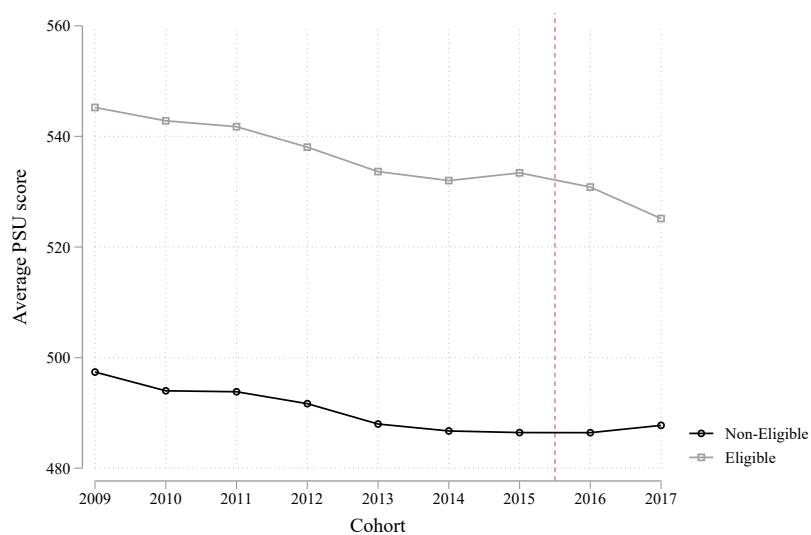
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Appendices

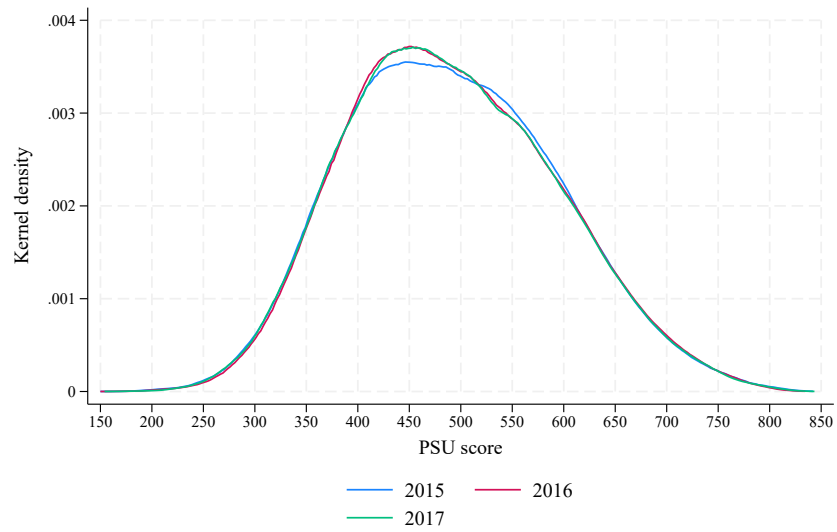
A Figures

Figure A1: Average PSU score over time for participating and non-participating institutions



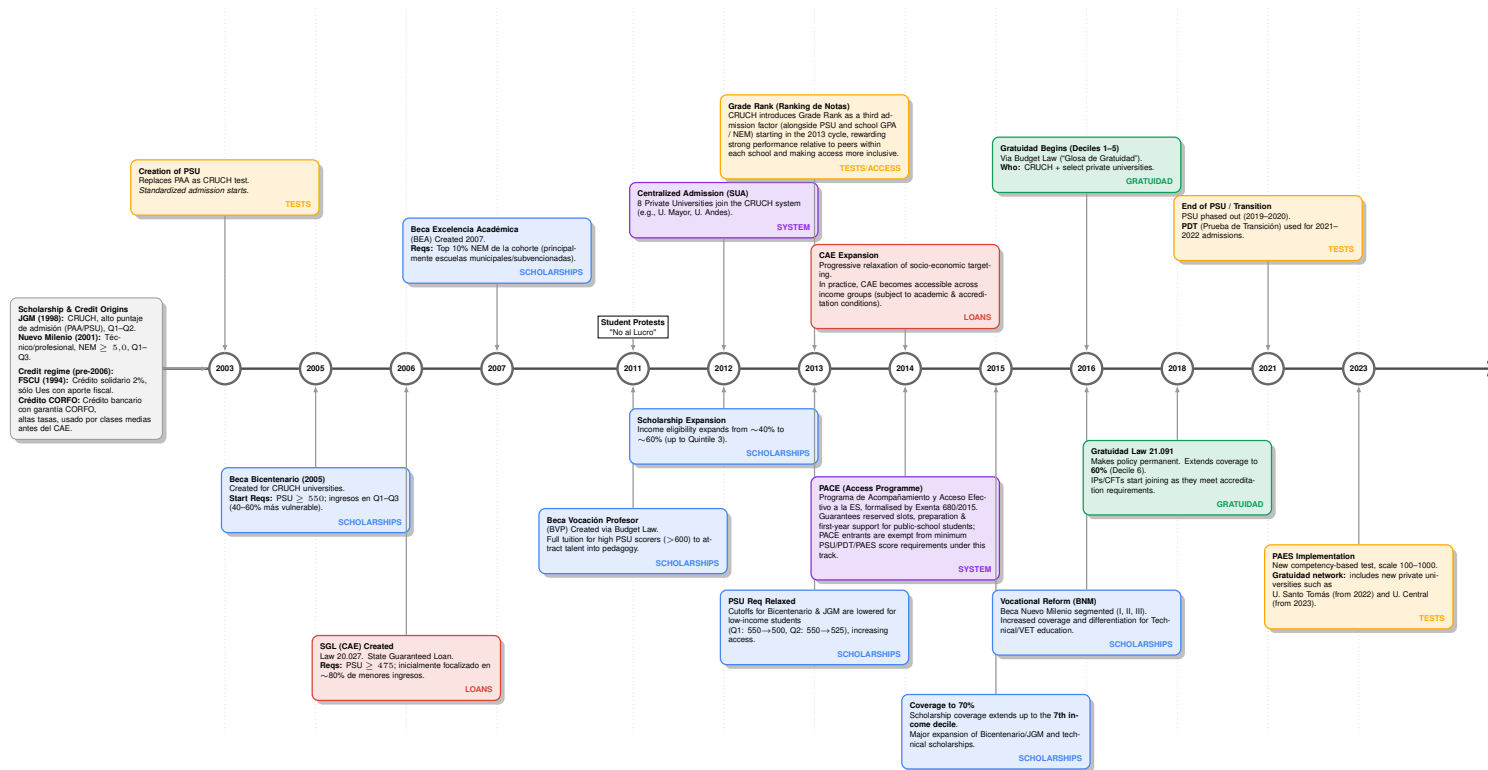
Notes: Annual average PSU score by groups of eligible and non-eligible institutions to free higher education.

Figure A2: Average PSU score over time for participating and non-participating institutions



Notes: Kernel density of PSU average scores enrolled in higher education for years 2015, 2016 and 2017.

Figure A3: Education reforms 2003-2024



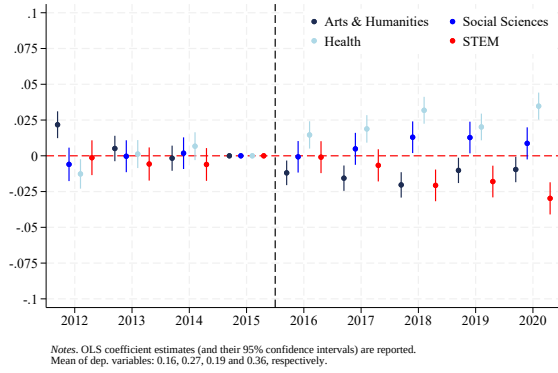
	PSU $\geq$ 510	475 $\leq$ PSU < 510	PSU < 475
Deciles 1-5	Scholarship TFHE (univ) TFHE (univ+SCP)	SGL loan TFHE (univ) TFHE (univ+SCP)	No public aid TFHE (univ) TFHE (univ+SCP)
Deciles 6-7	Scholarship	SGL loan	No public aid
Deciles 8-10	SGL loan	SGL loan	No public aid

PSU = admissions test; SGL = State-Guaranteed Loan; SCP = short-cycle programs; TFHE = Tuition-Free Higher Education

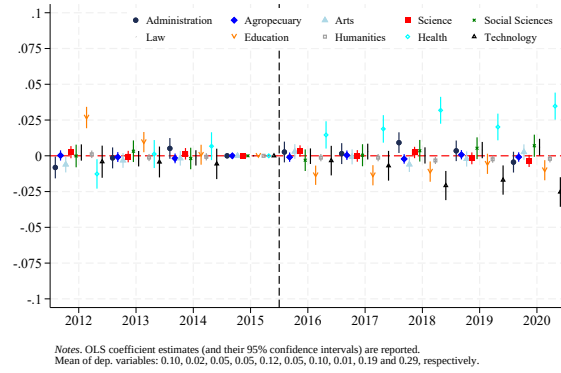
Years: 2015 2016 2017

Figure A4: Changes in funding instruments after TFHE implementation

Figure A5: DID: Preferences



a) 4 fields



b) 10 fields

Notes: These figures show the results of estimating Equation 1. Panel (a) groups the results in four fields, while panel (b) uses the original ten-field classification. STEM includes Agropecuaria, Science and Technology. Social Sciences includes Administration, Social Sciences nad Law. Arts and Humanities includes Education, Arts and Humanities. Health includes Health Sciences.

B Tables

Table B1: Descriptive statistics, students

	2012	2013	2014	2015	2016	2017
Enrollment						
N Students	165762	169257	169415	176027	180128	180302
Enrolled in platform	.260	.273	.280	.275	.280	.276
Enrolled out of platform	.338	.351	.346	.341	.330	.329
Not enrolled	.400	.374	.372	.382	.388	.394
Demographics						
Family Income	3.5	3.7	3.9	4	4.1	4.5
Private School	.111	.111	.111	.109	.106	.105
Private Health	.268	.267	.268	.263	.262	.235
Father With College	.168	.169	.170	.171	.169	.184
Mother Employed	.414	.436	.460	.460	.461	.461
Test Score	490	491	491	492	492	491
Free College	0	0	0	0	.145	.243
Subsidized Loan	.222	.213	.215	.187	.145	.105
Merit-based Scholarship	.136	.191	.200	.249	.158	.101
Field						
Business	.136	.136	.139	.148	.155	.160
Farming	.021	.022	.021	.023	.024	.025
Art and Architecture	.051	.050	.051	.052	.053	.055
Basic Sciences	.032	.033	.035	.033	.034	.031
Social Sciences	.081	.077	.076	.078	.083	.083
Law	.038	.035	.037	.037	.040	.040
Education	.110	.096	.091	.091	.095	.095
Humanities	.010	.010	.010	.010	.010	.009
Health	.214	.201	.200	.196	.196	.196
Technology	.284	.318	.318	.312	.290	.289

Notes: This table shows descriptive statistics on every student who enrolled and took the college entrance exam. Family income is categorized in 1-10 brackets, and field classification is performed following the ISCED-UNESCO guidelines.

	dropout	
OP fees	0.101***	(0.006)
Rel. Ability	-0.105***	(0.023)
Diff. Region	0.106***	(0.026)
Credits left	-0.113***	(0.004)
Designed total credits	0.099***	(0.006)
Delay	1.098***	(0.051)
Performance: category 0	2.116***	(0.040)
Performance: category 1	1.435***	(0.040)
Performance: category 2	0.291***	(0.033)
Performance: category 4	-0.721***	(0.049)
female	0.197**	(0.079)
Middle SES	0.033	(0.098)
High SES	0.007	(0.100)
PSU	0.063	(0.066)
GPA	-0.111**	(0.053)
Type 2	-10.276***	(0.656)
OP $\times$ female	-0.010**	(0.005)
OP $\times$ middle SES	-0.030***	(0.007)
OP $\times$ high SES	-0.027***	(0.008)
OP $\times$ psu	-0.026***	(0.004)
OP $\times$ gpa	0.015***	(0.003)
Delay $\times$ female	0.057	(0.050)
Delay $\times$ middle SES	0.040	(0.062)
Delay $\times$ high SES	0.233***	(0.062)
Delay $\times$ PSU	-0.191***	(0.035)
Delay $\times$ GPA	-0.062*	(0.034)
Delay $\times$ SCP	0.584***	(0.042)
t=2 $\times$ SCP	0.446***	(0.030)
t=3 $\times$ SCP	0.832***	(0.047)
t=4 $\times$ SCP	0.988***	(0.087)
t=5 $\times$ SCP	0.830***	(0.184)
t=6 $\times$ SCP	0.910**	(0.391)
Performance 0 $\times$ female	-0.301***	(0.039)
Performance 0 $\times$ middle SES	-0.060	(0.046)
Performance 0 $\times$ high SES	-0.093*	(0.051)
Performance 0 $\times$ PSU	-0.022	(0.027)
Performance 0 $\times$ GPA	-0.011	(0.026)
Performance 1 $\times$ female	-0.278***	(0.041)
Performance 1 $\times$ middle SES	-0.024	(0.048)
Performance 1 $\times$ high SES	0.053	(0.053)
Performance 1 $\times$ PSU	0.005	(0.027)
Performance 1 $\times$ GPA	0.072***	(0.027)
Performance 2 $\times$ female	-0.099***	(0.035)
Performance 2 $\times$ middle SES	-0.028	(0.043)
Performance 2 $\times$ high SES	0.082*	(0.046)
Performance 2 $\times$ PSU	-0.103***	(0.024)

Performance 2 $\times$ GPA	0.079***	(0.024)
Performance 4 $\times$ female	-0.328***	(0.052)
Performance 4 $\times$ middle SES	-0.046	(0.065)
Performance 4 $\times$ high SES	0.263***	(0.065)
Performance 4 $\times$ PSU	-0.180***	(0.036)
Performance 4 $\times$ GPA	-0.091**	(0.035)
Credits left $\times$ female	0.001	(0.002)
Credits left $\times$ middle SES	0.001	(0.003)
Credits left $\times$ high SES	-0.007**	(0.003)
Credits left $\times$ psu	0.007***	(0.002)
Credits left $\times$ gpa	0.010***	(0.002)
Rel. Abil. $\times$ female	0.038***	(0.011)
Rel. Abil. $\times$ middle SES	0.011	(0.013)
Rel. Abil. $\times$ high SES	0.057***	(0.015)
Rel. Abil. $\times$ psu	0.011*	(0.006)
Rel. Abil. $\times$ gpa	0.029***	(0.007)
Diff. Region $\times$ female	0.009	(0.027)
Diff. Region $\times$ middle SES	0.021	(0.034)
Diff. Region $\times$ high SES	0.037	(0.034)
Diff. Region $\times$ psu	0.034*	(0.018)
Diff. Region $\times$ gpa	-0.055***	(0.018)
Agriculture $\times$ fem	-0.073	(0.089)
Agriculture $\times$ middle SES	-0.034	(0.110)
Agriculture $\times$ high SES	-0.403***	(0.107)
Agriculture $\times$ psu	0.094	(0.069)
Agriculture $\times$ gpa	0.087	(0.059)
Agriculture $\times$ Type 2	0.055	(0.104)
Arts $\times$ fem	0.167**	(0.067)
Arts $\times$ middle SES	-0.011	(0.086)
Arts $\times$ high SES	-0.114	(0.080)
Arts $\times$ psu	0.077	(0.053)
Arts $\times$ gpa	-0.005	(0.046)
Arts $\times$ Type 2	-0.069	(0.084)
Science $\times$ fem	-0.047	(0.063)
Science $\times$ middle SES	-0.122	(0.081)
Science $\times$ high SES	0.064	(0.076)
Science $\times$ psu	-0.067	(0.050)
Science $\times$ gpa	0.226***	(0.043)
Science $\times$ Type 2	-1.016***	(0.084)
Social Sciences $\times$ fem	0.009	(0.055)
Social Sciences $\times$ middle SES	-0.039	(0.070)
Social Sciences $\times$ high SES	0.062	(0.066)
Social Sciences $\times$ psu	0.243***	(0.042)
Social Sciences $\times$ gpa	0.112***	(0.037)
Social Sciences $\times$ Type 2	0.014	(0.072)
Law $\times$ fem	-0.068	(0.060)
Law $\times$ middle SES	0.023	(0.076)

Law $\times$ high SES	-0.109	(0.071)
Law $\times$ psu	0.161***	(0.047)
Law $\times$ gpa	0.025	(0.040)
Law $\times$ Type 2	-0.536***	(0.074)
Education $\times$ fem	0.212***	(0.057)
Education $\times$ middle SES	-0.020	(0.067)
Education $\times$ high SES	-0.010	(0.071)
Education $\times$ psu	0.038	(0.041)
Education $\times$ gpa	0.076**	(0.037)
Education $\times$ Type 2	-0.408***	(0.069)
Humanities $\times$ fem	-0.036	(0.132)
Humanities $\times$ middle SES	-0.191	(0.141)
Humanities $\times$ high SES	-0.059	(0.152)
Humanities $\times$ psu	0.194**	(0.089)
Humanities $\times$ gpa	0.067	(0.081)
Humanities $\times$ Type 2	-0.451***	(0.170)
SCP $\times$ fem	0.053	(0.092)
SCP $\times$ middle SES	-0.154	(0.108)
SCP $\times$ high SES	-0.231*	(0.134)
SCP $\times$ psu	0.321***	(0.067)
SCP $\times$ gpa	0.064	(0.059)
SCP $\times$ type 2	1.215***	(0.128)
SCP-Gratuidad $\times$ fem	0.263***	(0.049)
SCP-Gratuidad $\times$ middle SES	-0.020	(0.060)
SCP-Gratuidad $\times$ high SES	-0.243***	(0.064)
SCP-Gratuidad $\times$ psu	0.408***	(0.039)
SCP-Gratuidad $\times$ gpa	0.215***	(0.033)
SCP-Gratuidad $\times$ type 2	0.796***	(0.073)
Health $\times$ fem	-0.024	(0.053)
Health $\times$ middle SES	-0.025	(0.064)
Health $\times$ high SES	-0.146**	(0.063)
Health $\times$ psu	0.133***	(0.040)
Health $\times$ gpa	-0.074**	(0.034)
Health $\times$ type 2	0.291***	(0.065)
Technology $\times$ fem	-0.078	(0.048)
Technology $\times$ middle SES	-0.026	(0.059)
Technology $\times$ high SES	-0.194***	(0.057)
Technology $\times$ psu	0.318***	(0.036)
Technology $\times$ gpa	0.141***	(0.031)
Technology $\times$ type 2	-0.417***	(0.057)
OP $\times$ Type 2	0.034***	(0.006)
Delay $\times$ Type 2	4.829***	(0.621)
Performance 0 $\times$ Type 2	4.638***	(0.191)
Performance 1 $\times$ Type 2	3.665***	(0.191)
Performance 2 $\times$ Type 2	3.259***	(0.190)
Performance 4 $\times$ Type 2	3.594***	(0.295)
Credits left $\times$ Type 2	0.170***	(0.005)

Rel. Abil. x Type 2	0.042***	(0.014)
Dist. x Type 2	-0.022	(0.034)
t= 2	-0.882***	(0.021)
t= 3	-1.523***	(0.030)
t= 4	-1.734***	(0.041)
t= 5	-1.839***	(0.053)
t= 6	-1.782***	(0.062)
Agricultural	-0.588***	(0.092)
Arts and Architecture	-0.026	(0.072)
Science	0.801***	(0.070)
Social Sciences	-0.010	(0.058)
Law	-0.100	(0.065)
Education	0.252***	(0.057)
Humanities	0.674***	(0.135)
SCP	0.933***	(0.134)
SCP-Gratuidad	0.440***	(0.105)
Health	-0.531***	(0.057)
Technology	-0.196***	(0.048)
mineduc=3	-0.233***	(0.050)
mineduc=10	-0.276***	(0.054)
mineduc=20	0.897***	(0.063)
mineduc=31	-0.038	(0.063)
mineduc=39	-0.168***	(0.058)
mineduc=45	-0.259***	(0.052)
mineduc=54	0.415***	(0.081)
mineduc=71	-0.084*	(0.048)
mineduc=72	-0.311***	(0.052)
mineduc=75	-0.347***	(0.059)
mineduc=76	-0.393***	(0.054)
mineduc=78	-0.528***	(0.055)
mineduc=85	-0.289***	(0.062)
mineduc=86	-0.407***	(0.036)
mineduc=87	-0.584***	(0.042)
mineduc=88	-0.348***	(0.042)
mineduc=89	-0.711***	(0.044)
mineduc=90	-0.886***	(0.050)
mineduc=93	-0.909***	(0.068)
mineduc=94	-0.303***	(0.062)
mineduc=990	-0.168***	(0.052)
mineduc=991	0.169***	(0.048)
mineduc=992	-0.410***	(0.050)
mineduc=993	-0.448***	(0.021)
Observations	457,037	
Mean of Dep. Variable	0.205	
Standard errors in parentheses		

Logistic regression of choosing the outside option (dropping out) in period t . Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardised. Base category is low SES, mean ability, enrolled in Business at Universidad de Chile.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table B3: Performance ordered logit

	index	
female	-0.052	(0.052)
Middle SES	0.016	(0.062)
High SES	-0.180***	(0.065)
PSU	0.116***	(0.044)
GPA	0.173***	(0.035)
Type 2	-3.534***	(0.102)
OP fees	-0.042***	(0.004)
Rel. Ability	0.038***	(0.014)
Diff. Region	-0.039**	(0.017)
Credits left (t-1)	0.152***	(0.004)
Designed total credits	-0.170***	(0.004)
Delay (t-1)	-1.694***	(0.029)
Performance (t-1): category 0	-1.128***	(0.040)
Performance (t-1): category 1	-0.839***	(0.041)
Performance (t-1): category 2	-0.257***	(0.028)
Performance (t-1): category 4	0.846***	(0.033)
OP × female	0.008***	(0.003)
OP × middle SES	0.031***	(0.004)
OP × high SES	0.037***	(0.005)
OP × psu	-0.006**	(0.002)
OP × gpa	0.004**	(0.002)
Delay (t-1) × female	-0.165***	(0.025)
Delay (t-1) × middle SES	-0.003	(0.030)
Delay (t-1) × high SES	0.021	(0.031)
Delay (t-1) × psu	0.156***	(0.019)
Delay (t-1) × gpa	0.008	(0.017)
Delay (t-1) × scp	0.455***	(0.033)
t = 2 × SCP	-0.489***	(0.026)
t = 3 × SCP	-1.680***	(0.042)
t = 4 × SCP	-3.042***	(0.071)
t = 5 × SCP	-3.302***	(0.146)
t = 6 × SCP	-2.925***	(0.325)
t = 7 × SCP	-1.733***	(0.670)
Performance (t-1) 0 × female	-0.101***	(0.034)
Performance (t-1) 0 × middle SES	-0.002	(0.041)
Performance (t-1) 0 × high SES	-0.069	(0.043)
Performance (t-1) 0 × PSU	0.077***	(0.025)
Performance (t-1) 0 × GPA	0.108***	(0.023)
Performance (t-1) 1 × female	0.172***	(0.043)
Performance (t-1) 1 × middle SES	0.061	(0.050)
Performance (t-1) 1 × high SES	0.094*	(0.054)
Performance (t-1) 1 × PSU	-0.281***	(0.029)
Performance (t-1) 1 × GPA	0.043	(0.028)
Performance (t-1) 2 × female	0.116***	(0.030)
Performance (t-1) 2 × middle SES	-0.009	(0.036)
Performance (t-1) 2 × high SES	0.055	(0.038)
Performance (t-1) 2 × PSU	-0.150***	(0.021)
Performance (t-1) 2 × GPA	0.089***	(0.020)
Performance (t-1) 4 × female	0.043	(0.034)
Performance (t-1) 4 × middle SES	-0.032	(0.040)
Performance (t-1) 4 × high SES	0.026	(0.042)
Performance (t-1) 4 × PSU	-0.069***	(0.024)
Performance (t-1) 4 × GPA	0.130***	(0.022)
Credits left (t-1) × female	0.025***	(0.002)
Credits left (t-1) × middle SES	0.005**	(0.002)
Credits left (t-1) × high SES	0.012***	(0.002)
Credits left (t-1) × psu	0.010***	(0.001)
Credits left (t-1) × gpa	0.006***	(0.001)
Rel. Abil. × female	-0.004	(0.007)
Rel. Abil. × middle SES	-0.001	(0.008)
Rel. Abil. × high SES	0.003	(0.010)
Rel. Abil. × psu	-0.019***	(0.004)
Rel. Abil. × gpa	0.035***	(0.004)
Diff. Region × female	-0.034**	(0.017)
Diff. Region × middle SES	0.004	(0.021)
Diff. Region × high SES	-0.043**	(0.021)
Diff. Region × psu	-0.007	(0.012)
Diff. Region × gpa	0.019*	(0.012)
OP × Type 2	-0.009**	(0.004)
Delay (t-1) × Type 2	1.069***	(0.048)
Performance (t-1) 0 × Type 2	-0.249***	(0.067)
Performance (t-1) 1 × Type 2	-0.062	(0.064)
Performance (t-1) 2 × Type 2	-0.207***	(0.056)
Performance (t-1) 4 × Type 2	-0.336***	(0.074)
Credits left (t-1) × Type 2	0.009**	(0.004)
Rel. Abil. × Type 2	0.044***	(0.009)
Dist. × Type 2	-0.090***	(0.023)
t = 2	-1.031***	(0.030)
t = 3	-0.079**	(0.035)
t = 4	0.999***	(0.041)
t = 5	1.558***	(0.049)
t = 6	1.100***	(0.057)
t = 7	0.390***	(0.063)
Observations	553,563	
Mean of Dep. Variable	3.113	

	index	
female	-0.052	(0.052)
Middle SES	0.016	(0.062)
High SES	-0.180***	(0.065)
PSU	0.116***	(0.044)
GPA	0.173***	(0.035)
Type 2	-3.534***	(0.102)
OP fees	-0.042***	(0.004)
Rel. Ability	0.038***	(0.014)
Diff. Region	-0.039**	(0.017)
Credits left (t-1)	0.152***	(0.004)
Designed total credits	-0.170***	(0.004)
Delay (t-1)	-1.694***	(0.029)
Performance (t-1): category 0	-1.128***	(0.040)
Performance (t-1): category 1	-0.839***	(0.041)
Performance (t-1): category 2	-0.257***	(0.028)
Performance (t-1): category 4	0.846***	(0.033)
OP $\times$ female	0.008***	(0.003)
OP $\times$ middle SES	0.031***	(0.004)
OP $\times$ high SES	0.037***	(0.005)
OP $\times$ psu	-0.006**	(0.002)
OP $\times$ gpa	0.004**	(0.002)
Delay (t-1) $\times$ female	-0.165***	(0.025)
Delay (t-1) $\times$ middle SES	-0.003	(0.030)
Delay (t-1) $\times$ high SES	0.021	(0.031)
Delay (t-1) $\times$ psu	0.156***	(0.019)
Delay (t-1) $\times$ gpa	0.008	(0.017)
Delay (t-1) $\times$ scp	0.455***	(0.033)
t = 2 $\times$ SCP	-0.489***	(0.026)
t = 3 $\times$ SCP	-1.680***	(0.042)
t = 4 $\times$ SCP	-3.042***	(0.071)
t = 5 $\times$ SCP	-3.302***	(0.146)
t = 6 $\times$ SCP	-2.925***	(0.325)
t = 7 $\times$ SCP	-1.733***	(0.670)
Performance (t-1) 0 $\times$ female	-0.101***	(0.034)
Performance (t-1) 0 $\times$ middle SES	-0.002	(0.041)
Performance (t-1) 0 $\times$ high SES	-0.069	(0.043)
Performance (t-1) 0 $\times$ PSU	0.077***	(0.025)
Performance (t-1) 0 $\times$ GPA	0.108***	(0.023)
Performance (t-1) 1 $\times$ female	0.172***	(0.043)
Performance (t-1) 1 $\times$ middle SES	0.061	(0.050)
Performance (t-1) 1 $\times$ high SES	0.094*	(0.054)
Performance (t-1) 1 $\times$ PSU	-0.281***	(0.029)
Performance (t-1) 1 $\times$ GPA	0.043	(0.028)
Performance (t-1) 2 $\times$ female	0.116***	(0.030)
Performance (t-1) 2 $\times$ middle SES	-0.009	(0.036)
Performance (t-1) 2 $\times$ high SES	0.055	(0.038)

Performance (t-1) 2 × PSU	-0.150***	(0.021)
Performance (t-1) 2 × GPA	0.089***	(0.020)
Performance (t-1) 4 × female	0.043	(0.034)
Performance (t-1) 4 × middle SES	-0.032	(0.040)
Performance (t-1) 4 × high SES	0.026	(0.042)
Performance (t-1) 4 × PSU	-0.069***	(0.024)
Performance (t-1) 4 × GPA	0.130***	(0.022)
Credits left (t-1) × female	0.025***	(0.002)
Credits left (t-1) × middle SES	0.005**	(0.002)
Credits left (t-1) × high SES	0.012***	(0.002)
Credits left (t-1) × psu	0.010***	(0.001)
Credits left (t-1) × gpa	0.006***	(0.001)
Rel. Abil. × female	-0.004	(0.007)
Rel. Abil. × middle SES	-0.001	(0.008)
Rel. Abil. × high SES	0.003	(0.010)
Rel. Abil. × psu	-0.019***	(0.004)
Rel. Abil. × gpa	0.035***	(0.004)
Diff. Region × female	-0.034**	(0.017)
Diff. Region × middle SES	0.004	(0.021)
Diff. Region × high SES	-0.043**	(0.021)
Diff. Region × psu	-0.007	(0.012)
Diff. Region × gpa	0.019*	(0.012)
Agriculture × fem	-0.246***	(0.048)
Agriculture × middle SES	-0.009	(0.059)
Agriculture × high SES	0.016	(0.057)
Agriculture × psu	-0.193***	(0.039)
Agriculture × gpa	-0.126***	(0.033)
Agriculture × Type 2	0.306***	(0.069)
Arts × fem	-0.021	(0.042)
Arts × middle SES	-0.024	(0.054)
Arts × high SES	0.139***	(0.051)
Arts × psu	-0.140***	(0.035)
Arts × gpa	-0.153***	(0.029)
Arts × Type 2	-0.360***	(0.056)
Science × fem	-0.241***	(0.042)
Science × middle SES	0.019	(0.053)
Science × high SES	0.032	(0.051)
Science × psu	-0.038	(0.034)
Science × gpa	-0.205***	(0.029)
Science × Type 2	0.251***	(0.056)
Social Sciences × fem	0.055	(0.036)
Social Sciences × middle SES	-0.026	(0.044)
Social Sciences × high SES	-0.059	(0.043)
Social Sciences × psu	-0.186***	(0.027)
Social Sciences × gpa	-0.106***	(0.024)
Social Sciences × Type 2	-0.316***	(0.047)
Law × fem	-0.207***	(0.037)

Law $\times$ middle SES	-0.074	(0.046)
Law $\times$ high SES	0.103**	(0.045)
Law $\times$ psu	-0.072**	(0.029)
Law $\times$ gpa	-0.107***	(0.025)
Law $\times$ Type 2	0.220***	(0.051)
Education $\times$ fem	0.075**	(0.037)
Education $\times$ middle SES	-0.005	(0.043)
Education $\times$ high SES	0.116**	(0.048)
Education $\times$ psu	-0.111***	(0.028)
Education $\times$ gpa	-0.099***	(0.024)
Education $\times$ Type 2	-0.258***	(0.047)
Humanities $\times$ fem	-0.006	(0.096)
Humanities $\times$ middle SES	-0.048	(0.099)
Humanities $\times$ high SES	0.097	(0.111)
Humanities $\times$ psu	-0.047	(0.065)
Humanities $\times$ gpa	-0.184***	(0.059)
Humanities $\times$ Type 2	0.104	(0.114)
SCP $\times$ fem	0.209***	(0.063)
SCP $\times$ middle SES	-0.030	(0.073)
SCP $\times$ high SES	0.297***	(0.095)
SCP $\times$ psu	0.058	(0.046)
SCP $\times$ gpa	-0.067	(0.041)
SCP $\times$ type 2	-0.625***	(0.077)
SCP-Gratuidad $\times$ fem	0.283***	(0.034)
SCP-Gratuidad $\times$ middle SES	0.057	(0.040)
SCP-Gratuidad $\times$ high SES	0.184***	(0.045)
SCP-Gratuidad $\times$ psu	0.072***	(0.027)
SCP-Gratuidad $\times$ gpa	0.055**	(0.023)
SCP-Gratuidad $\times$ type 2	-0.356***	(0.055)
Health $\times$ fem	-0.066**	(0.031)
Health $\times$ middle SES	-0.042	(0.037)
Health $\times$ high SES	0.113***	(0.037)
Health $\times$ psu	-0.000	(0.024)
Health $\times$ gpa	-0.117***	(0.020)
Health $\times$ type 2	-0.048	(0.043)
Technology $\times$ fem	-0.136***	(0.030)
Technology $\times$ middle SES	-0.039	(0.035)
Technology $\times$ high SES	0.038	(0.035)
Technology $\times$ psu	0.007	(0.023)
Technology $\times$ gpa	-0.061***	(0.019)
Technology $\times$ type 2	0.205***	(0.039)
OP $\times$ Type 2	-0.009**	(0.004)
Delay (t-1) $\times$ Type 2	1.069***	(0.048)
Performance (t-1) 0 $\times$ Type 2	-0.249***	(0.067)
Performance (t-1) 1 $\times$ Type 2	-0.062	(0.064)
Performance (t-1) 2 $\times$ Type 2	-0.207***	(0.056)
Performance (t-1) 4 $\times$ Type 2	-0.336***	(0.074)

Credits left (t-1) $\times$ Type 2	0.009**	(0.004)
Rel. Abil. $\times$ Type 2	0.044***	(0.009)
Dist. $\times$ Type 2	-0.090***	(0.023)
t = 2	-1.031***	(0.030)
t = 3	-0.079**	(0.035)
t = 4	0.999***	(0.041)
t = 5	1.558***	(0.049)
t = 6	1.100***	(0.057)
t = 7	0.390***	(0.063)
Agricultural	-0.077	(0.049)
Arts and Architecture	0.074*	(0.045)
Science	-0.606***	(0.047)
Social Sciences	0.395***	(0.037)
Law	-0.264***	(0.040)
Education	0.341***	(0.036)
Humanities	-0.117	(0.099)
SCP	0.880***	(0.090)
SCP-Gratuidad	1.014***	(0.067)
Health	-0.063*	(0.033)
Technology	-0.326***	(0.030)
mineduc=3	0.166***	(0.031)
mineduc=10	0.499***	(0.034)
mineduc=20	-0.715***	(0.044)
mineduc=31	0.508***	(0.040)
mineduc=39	0.505***	(0.037)
mineduc=45	0.237***	(0.033)
mineduc=54	0.702***	(0.053)
mineduc=71	0.132***	(0.031)
mineduc=72	0.261***	(0.032)
mineduc=75	0.214***	(0.037)
mineduc=76	0.179***	(0.033)
mineduc=78	0.116***	(0.032)
mineduc=85	0.451***	(0.039)
mineduc=86	-0.002	(0.023)
mineduc=87	0.159***	(0.026)
mineduc=88	-0.293***	(0.027)
mineduc=89	-0.005	(0.027)
mineduc=90	0.029	(0.030)
mineduc=93	0.121***	(0.039)
mineduc=94	0.064	(0.039)
mineduc=990	0.094***	(0.034)
mineduc=991	0.457***	(0.032)
mineduc=992	-0.326***	(0.032)
mineduc=993	0.146***	(0.014)
cut1	-4.826***	(0.084)
cut2	-3.994***	(0.084)
cut3	-2.613***	(0.084)

cut4	-2.111***	(0.084)
Observations	553,563	
Mean of Dep. Variable	3.113	

Standard errors in parentheses

Ordered logistic regression of discretized credit achievement (0-4). Ordered logistic regression of discretized credit achievement (0-4). Controls include vector of characteristics, Institution and major FE, year enrolled FE and cumulated performance. PSU and GPU are standardized. Base category is low SES, mean ability, enrolled in Business at Universidad de Chile.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

C Estimation

C.1 Classification of unobserved types

We classify enrolled students into M latent types via k-means on two standardized, individual-average residual moments: completion conditional on registered credits and completion relative to the intended annual course load. Both moments are obtained as OLS residuals from

$$y_{it} = \mathbf{x}_i' \alpha_x + \mathbf{z}_{it}' \alpha_z + \lambda_{field(j)} + \lambda_{inst(j)} + \lambda_t + \varepsilon_{it}, \quad (21)$$

where y_{it} is, in turn, the registered-credit completion rate and the intended-annual-load completion rate, $\mathbf{x}_i$ collects individual controls (sex, standardized PSU and GPA, socioeconomic status), and $\mathbf{z}_{it}$ collects state controls (previous-period and cumulative credit completion, delay, and remaining-credits-to-graduate indicators). Fixed effects for field, institution (HEI), and period are included, allowed to differ between SCP and university programs. Residuals are collapsed to individual-level averages and standardized to unit variance before clustering.

Clustering uses k-means++ initialization: cluster centers are seeded sequentially with probability proportional to squared distance from existing centers, and the procedure is repeated over 20 independent seeds to guard against convergence to a local optimum, keeping the run with the lowest within-cluster sum of squares (WCSS). Resulting clusters are relabeled from highest to lowest average of the two standardized moments, so that Type 1 always denotes the highest-productivity group. $M = 2$ is our production classification, entering the structural estimation as an element of Z_i ; $M = 3$ is estimated on the same moments purely as a robustness check on this choice of M and is not used in any structural estimate.

Table B5 compares the fit and stability of the $M = 2$ and $M = 3$ classifications, and Table B6 compares the resulting type profiles.

Table B5: Comparison of two- and three-type classifications

K	Students	Moments	WCSS	Q(K)	Explained (%)	CH	Best seed	Optimum starts	Min. share (%)	Max. share (%)
2	193,066	2	130,669	0.677	66.2	377,445	1	20	25.0	75.0
3	193,066	2	80,678	0.418	79.1	365,474	2	4	10.2	65.9

Notes: Both classifications use the same two standardized individual-average residual moments and 20 k-means++ starts. WCSS mechanically decreases with K. K=2 is the production classification; K=3 is appendix-diagnostic only.

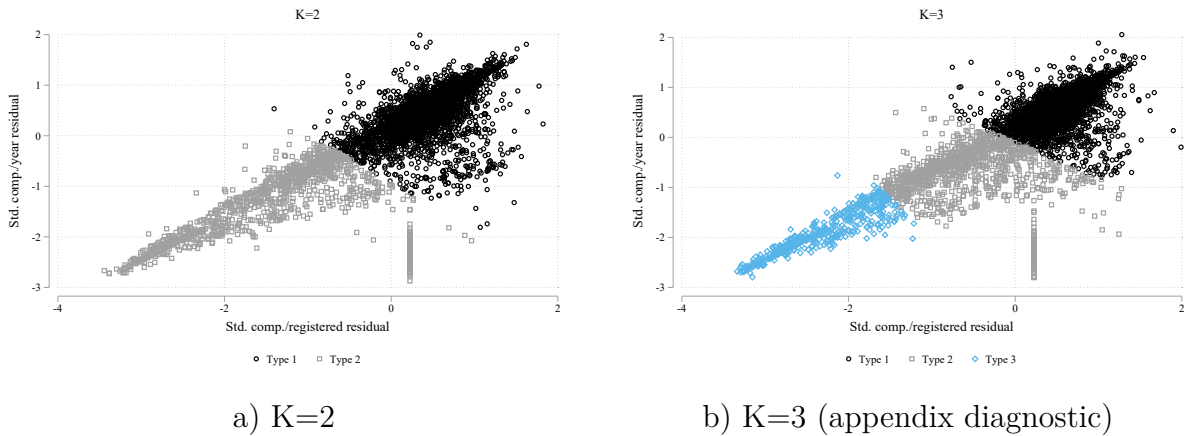
Moving from $M = 2$ to $M = 3$ mechanically lowers WCSS and raises the share of variance explained (Table B5), since additional clusters can only improve within-sample fit. The improvement is not merely mechanical curve-fitting, however: the Calinski-Harabasz index

Table B6: Profiles of two- and three-type classifications

Statistic	K=2		K=3		
	Type 1	Type 2	Type 1	Type 2	Type 3
Students	144,819	48,247	127,285	46,158	19,623
Share (%)	75.0	25.0	65.9	23.9	10.2
Std. comp./registered residual	0.459	-1.377	0.536	-0.479	-2.354
Std. comp./year residual	0.480	-1.441	0.596	-0.824	-1.928
Completed/registered ratio	0.901	0.299	0.929	0.563	0.108
Completed/intended-year ratio	0.840	0.209	0.882	0.388	0.080
Graduated by current credit state (%)	51.8	0.2	58.0	3.0	0.0
Behavioral dropout before t=6 (%)	38.6	99.3	32.4	93.2	100.0
Forced terminal exit at t=6 (%)	9.5	0.4	9.6	3.7	0.0
Observed academic periods	3.63	1.50	3.73	2.19	1.13

Notes: Types are ordered from highest to lowest mean standardized academic-productivity residual. Behavioral dropout excludes forced sixth-year nongraduate exit. K=2 is used in all structural estimates; K=3 is diagnostic only.

Figure A6: Classification moments by type: K=2 vs. K=3



Notes: Standardized comp./registered residual (x-axis) against standardized comp./year residual (y-axis), by assigned type. A random subsample of 5,000 students is plotted for legibility; the k-means fit itself uses the full classification sample.

also favors $M = 2$ over $M = 3$ once the loss of degrees of freedom is taken into account, and the additional cluster comes from splitting the low-productivity $M = 2$ group (Type 2, 25.1% of students) into a moderate-productivity group and a very small, very-low-productivity group (Type 3 under $M = 3$, 10.5% of students, completed/registered ratio of 0.107, and essentially universal behavioral dropout before the sixth period). This third group is small enough, and separated enough from the rest of the distribution, that Figure A6 shows the additional structure is visually apparent but concentrated in a thin tail rather than reflecting a qualitatively distinct third mode. We therefore retain $M = 2$ for parsimony and cluster-size balance in the production structural estimates, treating $M = 3$ as an appendix robustness check on the sensitivity of the type classification to M .

C.2 Functional forms

C.2.1 Performance outcomes

For $g \in \mathcal{G} = \{0, 1, 2, 3, 4\}$, denote $\tilde{g}$ the underlying index:

$$\begin{aligned} \tilde{g}_j(x_{it}) = & \sum_m \sum_n \alpha_{mn}^g Z_{im} Z_{jn} + \sum_m \sum_n \alpha_{gm}^g Z_{gm} Z_{in} + \mathbf{Z}_g' \alpha_G^g \\ & + \alpha_{field(j)}^g + \alpha_{inst(j)}^g + \lambda_t^g + \lambda_{tscp}^g \cdot \text{SCP}_{ij}, \end{aligned} \quad (22)$$

where $Z_i = (\text{psu}, \text{gpa}, \text{income}, \text{female})$, $Z_j = (\text{field1}, \dots, \text{field10})$, $Z_g = (g_{it-1}, G_{it-1}, \text{delay}_{it-1}, \text{1left}_{it-1}, \text{2left}_{it-1}, \text{3left}_{it-1})$. g_{it-1} and G_{it-1} denote previous-period and cumulative credit completion, respectively. Xleft indicates whether the student left the program with X or more credits remaining. $\alpha_{field(j)}^g$ are field fixed effects, $\alpha_{inst(j)}^g$ are institution fixed effects, and λ_t^g are period fixed effects. Period fixed effects are allowed to differ between SCPs and university programs.

Xleft variables take into account the fact that passing all the credits in a period where less credits are needed to graduate might be different. An alternative modelling approach would estimate separate models for each possible credit left in a year (1,2,3,4). However, estimating it together gives us more precise estimates.

C.2.2 Dropout probability

We estimate the following logistic regression by maximum likelihood:

$$\begin{aligned} \tilde{drop}_j(x_{it}) = & \sum_m \sum_n \alpha_{mn}^d Z_{im} Z_{jn} + \sum_m \sum_n \alpha_{gm}^d Z_{gm} Z_{in} + \mathbf{Z}_g' \alpha_G^d \\ & + \alpha_{field(j)}^d + \alpha_{inst(j)}^d + \lambda_t^d + \lambda_{tscp}^d \cdot \text{SCP}_{ij}, \end{aligned} \quad (23)$$

where $Z_i = (\text{psu}, \text{gpa}, \text{income}, \text{female})$, $Z_j = (\text{field1}, \dots, \text{field10})$, $Z_g = (g_{it-1}, G_{it-1}, \text{delay}_{it-1}, \text{1left}_{it-1}, \text{2left}_{it-1}, \text{3left}_{it-1})$. g_{it-1} and G_{it-1} denote previous-period and cumulative credit completion, respectively. X_{left} indicates whether the student left the program with X or more credits remaining. $\alpha_{\text{field}(j)}^g$ are field fixed effects, $\alpha_{\text{inst}(j)}^g$ are institution fixed effects, and λ_t^g are period fixed effects. Period fixed effects are allowed to differ between SCPs and university programs.

C.2.3 Labour market

We estimate the following wage equation using CASEN survey data,

$$\begin{aligned} \log(w_j(x_{it})) = & \beta_0^w + \beta_{f, \text{field}(j)}^w \text{female}_i + \beta_e^w \text{exper}_{it} + \beta_{e2}^w \text{exper}_{it}^2 \\ & + \sum_{r=1}^{16} \beta_{r, \text{field}(j)}^w R_{ir} + \beta_{\text{inst_type}(j), \text{field}(j)}^w + \varepsilon_{ijt}, \end{aligned} \quad (24)$$

where R_{ir} is a dummy for region r and $\text{inst_type}(j) \in \{\text{SCP}, \text{Private}, \text{CRUCH}\}$ denotes the institution type of program j . The gender gap and the institution-type premium are both allowed to vary by field. The region-field interactions $\{R_{ir} \beta_{r, \text{field}(j)}^w\}$ are crucial for identification of the wage coefficient α_w in the structural model, as they serve as an exclusion restriction: regional-field variation affects utility only through wages.

CASEN observes broad institution type but not the specific institution attended. To capture within-type variation in earnings, we separately regress log wages on institution (`hei_code`) and field fixed effects using institution-level graduate earnings data scraped from the Chilean Ministry of Education's [Mifuturo](#) platform. Because part of an institution's Mifuturo fixed effect already reflects its institution type, we demean each institution's fixed effect by subtracting the mean fixed effect across institutions of the same type, so that only the institution's deviation from its type average is retained. We then predict [Equation 24](#) for each student in the sample and add this demeaned institution premium, which avoids double-counting the type-level premium already identified in [Equation 24](#).

C.2.4 Fixed costs

$$\begin{aligned} FC_j(x_{it}) = & \alpha_p OP_{ij} + \sum_m \alpha_{pm} OP_{ij} Z_{im} + \sum_m \sum_n \alpha_{mn} Z_{im} Z_{jn} + \sum_m \sum_n \alpha_{gm} Z_{gm} Z_{in} \\ & + \mathbf{Z}_g' \alpha_G + \sum_m \sum_k \alpha_{km} M_{ij,k} Z_{im} + \mathbf{M}_{ij}' \alpha_m + \alpha_{\text{field}(j)} + \alpha_{\text{inst}(j)}, \end{aligned} \quad (25)$$

where D_{ij} is a dummy for the institution being in a different region (distance proxy), and A_{ij} is the student's relative ability in program j . $M_{ij} = (D_{ij}, A_{ij})$ is a vector of student-program

matched variables.

C.3 Conditional Value Function

Conditional value and the terminal boundary. Let $\mathcal{A}_{i,t+1}(\bar{g})$ indicate that, after performance outcome $\bar{g}$, the student has a genuine choice between staying and dropping out. The post-performance continuation value $\bar{V}(x_{i,t+1}(\bar{g}))$ is

$$\bar{V}(x_{i,t+1}(\bar{g})) = \begin{cases} \gamma + \alpha_w \text{wage}_0(x_{i,t+1}) - \ln p_{0,t+1}(x_{i,t+1}(\bar{g})), & \mathcal{A}_{i,t+1}(\bar{g}) = 1, \\ \alpha_w \text{wage}_j(x_{i,t+1}), & \text{graduation,} \\ \alpha_w \text{wage}_0(x_{i,t+1}), & \text{terminal nongraduation.} \end{cases} \quad (26)$$

The final academic period still contains effort and a performance draw. What is absent is a choice after that draw: a nongraduate exits mechanically. Thus the terminal nongraduation branch contains the dropout wage, but neither $-\ln p_0$ nor γ .

For an enrolled alternative,

$$v_j(x_{it}, e_{it}) = -FC_j(x_{it}) - c_j(x_{it})e_{it} + \beta \sum_{\bar{g} \in \mathcal{G}} \phi_{ijt}^{\bar{g}}(e_{it}) \bar{V}(x_{i,t+1}(\bar{g})). \quad (27)$$

At an interior effort optimum, the first-order condition implies

$$v_j(x_{it}, e_{ijt}^*) = -FC_j(x_{it}) + \beta \sum_{\bar{g} \in \mathcal{G}} a_{ijt}^{\bar{g}} \bar{V}(x_{i,t+1}(\bar{g})), \quad (28)$$

$$a_{ijt}^{\bar{g}} \equiv \phi_{ijt}^{\bar{g}}(e_{ijt}^*) - e_{ijt}^* \left. \frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}} \right|_{e_{it}=e_{ijt}^*}. \quad (29)$$

Separating the part whose coefficient is fixed at β from the wage component whose coefficient identifies α_w gives

$$EMAX_{ijt} = \sum_{\bar{g} \in \mathcal{G}} a_{ijt}^{\bar{g}} \mathbb{1}\{\mathcal{A}_{i,t+1}(\bar{g}) = 1\} [\gamma - \ln p_{0,t+1}(x_{i,t+1}(\bar{g}))], \quad (30)$$

$$EWAGE_{ijt} = \beta \sum_{\bar{g} \in \mathcal{G}} a_{ijt}^{\bar{g}} \text{wage}_{\bar{g}}(x_{i,t+1}), \quad (31)$$

$$v_j(x_{it}, e_{ijt}^*) = -FC_j(x_{it}) + \beta EMAX_{ijt} + \alpha_w EWAGE_{ijt}. \quad (32)$$

Here $\text{wage}_{\bar{g}} = \text{wage}_j$ on graduation branches and $\text{wage}_{\bar{g}} = \text{wage}_0$ otherwise. In the implementation's terminal period $t = 6$, the active-choice indicator is zero for every outcome, so $EMAX_{ijt} = 0$ while $EWAGE_{ijt}$ remains outcome-dependent. where $\Lambda(\cdot)$ denotes the standard logistic CDF. The general expression for ordered logit probabilities are, denoting

by $\alpha_{\bar{g}}$ the estimated cutoff points,

$$\begin{aligned}\phi_{ijt}^{\bar{g}}(e_j^*(x_{it})) &= \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \\ &= \frac{1}{1 + \exp(e_j^*(x_{it}) - \alpha_{\bar{g}})} - \frac{1}{1 + \exp(e_j^*(x_{it}) - \alpha_{\bar{g}-1})}\end{aligned}$$

The derivative wrt $e_j^*(x_{it})$ writes

$$\begin{aligned}\left. \frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}} \right|_{e_{it}=e_j^*(x_{it})} &= -\Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it}))) \\ &\quad - (-\Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})))) \\ \left. \frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}} \right|_{e_{it}=e_j^*(x_{it})} &= \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - e_j^*(x_{it}))) \\ &\quad - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - e_j^*(x_{it})))\end{aligned}$$

Computing the marginal-cost and envelope expression gives

$$\sum_{\bar{g} \in \mathcal{G}} a_{ijt}^{\bar{g}} \bar{V}(x_{i,t+1}(\bar{g})).$$

The ordered-logit expansions below describe the weights $a_{ijt}^{\bar{g}}$. Whenever they display the bracket $\alpha_w wage_0 - \ln p_0$, that bracket applies only to an active nongraduation branch. Graduation and terminal nongraduation instead use the wage-only branches in [Equation 26](#); their contribution to $EMAX$ is zero and their wage contribution remains in $EWAGE$.

For $g = 1$

$$\begin{aligned}(\Lambda(\alpha_1 - e_j^*(x_{it})) + (\Lambda(\alpha_1 - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_1 - e_j^*(x_{it})))) \times e_j^*(x_{it}) \\ (\Lambda(\alpha_1 - e_j^*(x_{it})) + (\Lambda(\alpha_1 - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_1 - e_j^*(x_{it})))) \times e_j^*(x_{it})\end{aligned}$$

$$\begin{aligned}(\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(1))) \times \\ \left(\Lambda(\alpha_1 - index_j(x_{it})) - \left. \frac{\partial \phi_{ijt}^{\bar{g}}(e_j(x_{it}))}{\partial e_{it}} \right|_{e_{it}=e_j^*(x_{it})} \times e_j^*(x_{it}) \right)\end{aligned}$$

We defined effort as

$$e_j(x_{it}) = \frac{1 - \Lambda(\alpha_1 - index_j(x_{it}))}{\Lambda(\alpha_1 - index_j(x_{it}))} \quad (33)$$

$$\Lambda(\alpha_1 - index_j(x_{it})) = \frac{1}{1 + e_j(x_{it})} \quad (34)$$

$$\alpha_1 - index_j(x_{it}) = \ln \left(\frac{\frac{1}{1+e_j(x_{it})}}{1 - \frac{1}{1+e_j(x_{it})}} \right) \quad (35)$$

$$\alpha_1 - index_j(x_{it}) = \ln \left(\frac{1}{e_j(x_{it})} \right) = -\ln(e_j(x_{it})) \quad (36)$$

$$\alpha_1 - \ln(e_j(x_{it})) = index_j(x_{it}) \quad (37)$$

then the derivative wrt effort looks like

$$\begin{aligned} & \left(\Lambda(\alpha_1 - index_j(x_{it})) - \left(\lambda(\alpha_1 - index_j(x_{it})) \times \frac{-1}{e_j(x_{it})} \right) \times e_j(x_{it}) \right) \\ &= \frac{1}{1 + e_j(x_{it})} + \frac{e_j(x_{it})}{(1 + e_j(x_{it}))^2} \end{aligned}$$

For $g \in (2, 4)$

$$\begin{aligned} & (\alpha_{wage_0}(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(\bar{g}))) \times \\ & (\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\ & (\Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it}))) - \Lambda(\alpha_{\bar{g}} - index_j(x_{it})) \times (1 - \Lambda(\alpha_{\bar{g}} - index_j(x_{it})))) \end{aligned}$$

Recall that rearranging the effort measure in [Equation 33](#), we can express $index_j(x_{it}) = \alpha_1 - \ln(e_j(x_{it}))$

$$\begin{aligned}
& (\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(\bar{g}))) \times \\
& (\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
& \left. \frac{\partial \phi_{ijt}^{\bar{g}}(e_j(x_{it}))}{\partial e_{it}} \Big|_{e_{it}=e_j^*(x_{it})} (\Lambda(\alpha_{\bar{g}} - \alpha_1 - \ln(e_j^*(x_{it}))) - \Lambda(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e_j^*(x_{it}))) \times e_j^*(x_{it})) \right) \\
& = (\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(\bar{g}))) \times \\
& (\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
& \left(\lambda(\alpha_{\bar{g}} - \alpha_1 - \ln(e_j^*(x_{it}))) \times \left(\frac{-1}{e_j^*(x_{it})} \right) - \lambda(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e_j^*(x_{it}))) \times \left(\frac{-1}{e_j^*(x_{it})} \right) \right) \times e_j^*(x_{it})) \\
& = (\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(\bar{g}))) \times \\
& (\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) - \\
& \left(\frac{-e_j^*(x_{it})}{e_j^*(x_{it})} \right) \times \left(\frac{\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e_j^*(x_{it})))}{(1 + \exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e_j^*(x_{it}))))^2} - \frac{\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e_j^*(x_{it})))}{(1 + \exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e_j^*(x_{it}))))^2} \right)) \\
& = (\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(\bar{g}))) \times \\
& (\Lambda(\alpha_{\bar{g}} - index_j(x_{it})) - \Lambda(\alpha_{\bar{g}-1} - index_j(x_{it})) + \\
& \left(\frac{\exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e_j^*(x_{it})))}{(1 + \exp(\alpha_{\bar{g}} - \alpha_1 - \ln(e_j^*(x_{it}))))^2} - \frac{\exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e_j^*(x_{it})))}{(1 + \exp(\alpha_{\bar{g}-1} - \alpha_1 - \ln(e_j^*(x_{it}))))^2} \right))
\end{aligned}$$

For $g = 5$

$$\begin{aligned}
& (\alpha_w wage_0(x_{it+1}) - \ln \Pr(d_{it+1}^0 | x_{it+1}(5))) \times \\
& (1 - \Lambda(\alpha_4 - e_j^*(x_{it})) - (\Lambda(\alpha_4 - e_j^*(x_{it})) \times (1 - \Lambda(\alpha_4 - e_j^*(x_{it})))) \times e_j^*(x_{it}))
\end{aligned}$$

$$\sum_{\bar{g} \in \mathcal{G}} \left(\phi_{ijt}^{\bar{g}}(e_j^*(x_{it})) - \frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}} \Big|_{e_{it}=e_j^*(x_{it})} e_j^*(x_{it}) \right)$$

Notice from the above expression that

$$\sum_{\bar{g} \in \mathcal{G}} (\phi_{ijt}^{\bar{g}}(e_j^*(x_{it}))) = 1$$

Since we are integrating over all possible states and that

$$\begin{aligned}
& \sum_{\bar{g} \in \mathcal{G}} \left(\frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}} \Big|_{e_{it}=e_j^*(x_{it})} e_j^*(x_{it}) \right) \\
&= \lambda(\alpha_1 - index(x_{it})) \times (-1/e_j^*(x_{it})) \times e_j^*(x_{it}) + \\
& (\lambda(\alpha_2 - index(x_{it})) - \lambda(\alpha_1 - index(x_{it}))) \times (-1/e_j^*(x_{it})) \times e_j^*(x_{it}) + \\
& (\lambda(\alpha_3 - index(x_{it})) - \lambda(\alpha_2 - index(x_{it}))) \times (-1/e_j^*(x_{it})) \times e_j^*(x_{it}) + \\
& (\lambda(\alpha_4 - index(x_{it})) - \lambda(\alpha_3 - index(x_{it}))) \times (-1/e_j^*(x_{it})) \times e_j^*(x_{it}) + \\
& (-\lambda(\alpha_4 - index(x_{it}))) \times (-1/e_j^*(x_{it})) \times e_j^*(x_{it})
\end{aligned}$$

since $index = \ln(e_j^*(x_{it})) + \alpha_1$ and that $\frac{\partial \lambda(\alpha_g - index(x_{it}))}{\partial e_{it}} = \lambda(\alpha_g - index(x_{it})) \times (-1/e_j^*(x_{it}))$.

This can be rewritten as

$$\begin{aligned}
& \sum_{\bar{g} \in \mathcal{G}} \left(\frac{\partial \phi_{ijt}^{\bar{g}}(e_{it})}{\partial e_{it}} \Big|_{e_{it}=e_j^*(x_{it})} e_j^*(x_{it}) \right) \\
&= -\lambda(\alpha_1 - index(x_{it})) + \\
& -(\lambda(\alpha_2 - index(x_{it})) - \lambda(\alpha_1 - index(x_{it}))) + \\
& -(\lambda(\alpha_3 - index(x_{it})) - \lambda(\alpha_2 - index(x_{it}))) + \\
& -(\lambda(\alpha_4 - index(x_{it})) - \lambda(\alpha_3 - index(x_{it}))) + \\
& -(-\lambda(\alpha_1 - index(x_{it}))) \\
&= 0
\end{aligned}$$

Period and alternative conventions. The outside option is terminal in every period:

$$v_0(x_{it}) = \alpha_w wage_0(x_{it}),$$

with no discount factor multiplying this contemporaneous terminal payoff. For an enrolled alternative before the terminal boundary, the active branches use the Hotz–Miller expression in [Equation 26](#). In the final academic period, effort still changes the probability of graduation, but the continuation value after performance is simply

$$\bar{V}(x_{i,T+1}(\bar{g})) = \begin{cases} \alpha_w wage_j(x_{i,T+1}), & \text{graduation,} \\ \alpha_w wage_0(x_{i,T+1}), & \text{nongraduation.} \end{cases}$$

Thus terminal *EMAX* is zero, whereas terminal *EWAGE* is generally nonzero.